

INDUSTRIAL NETWORKING
MENG 4010

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1.1 DATA COMMUNICATIONS

Communication systems are designed to transfer information from a source (sender) to a destination (receiver). The information, often referred to as data, consists of discrete symbols or encoded representations agreed upon by both sender and receiver.

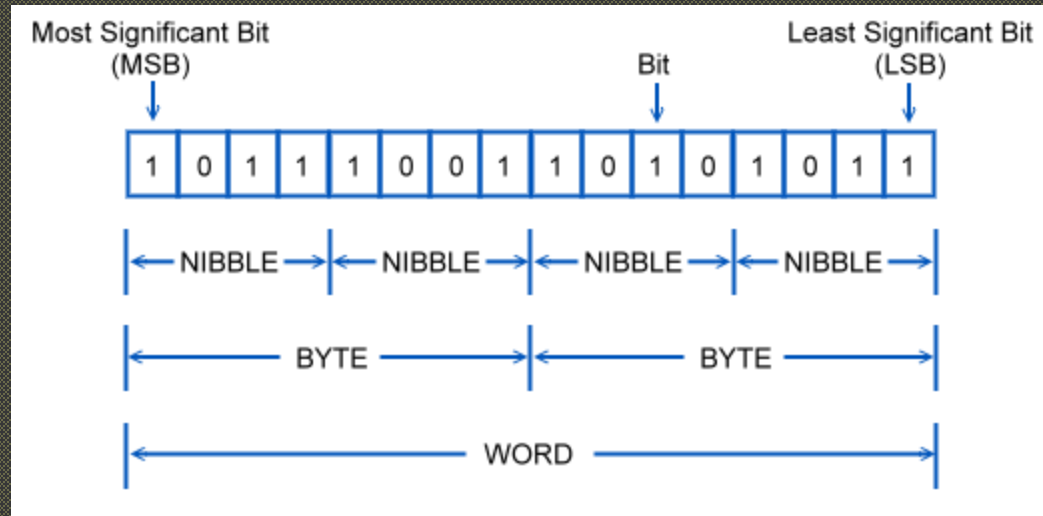
In digital systems (such as computers and PLCs), data is typically represented in binary form ("0" and "1"). These binary values are physically conveyed using different mediums:

- Electrical cables transmit data as voltage levels (e.g., high voltage = "1", low voltage = "0").
- Fiber-optic cables transmit data as light pulses (presence of light = "1", absence = "0").

ELECTRICAL CABLE (Voltage Levels)	FIBER-OPTIC CABLE (Light Pulses)
-----	-----
Binary Data: 0 1 0 1	Binary Data: 0 1 0 1
Signal: --- --- --- ---	Signal: --- --- --- ---
Voltage: LOW HIGH LOW HIGH	Light: OFF ON OFF ON

1.2 BIT, BYTE, WORD, CHARACTERS

Any number can be represented as a combination of “0s” and “1s”.



Decimal to Binary to Octal to HexaDecimal

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F
16	10000	20	10
17	10001	21	11
18	10010	22	12
19	10011	23	13
20	10100	24	14

<https://www.java67.com/2012/10/convert-hexadecimal-to-decimal-binary-octal-java-example.html>

1.2 BIT, BYTE, WORD, CHARACTERS

ASCII (American Standard Code for Information Interchange) is a character encoding standard that assigns numerical values to letters, digits, punctuation, and control codes.

1. ASCII Control Characters (0–31 & 127)

- Non-printable codes used for device control and text formatting.
- Examples:
 - NUL (0) – Null character (placeholder)
 - LF (10) – Line Feed (moves cursor down)
 - CR (13) – Carriage Return (returns cursor to start)
 - ESC (27) – Escape (initiates control sequences)
 - DEL (127) – Delete (removes a character)

2. ASCII Printable Characters (32–126)

- Visible symbols, including:
- Space (32) – Blank space
- Digits (48–57) – 0 to 9
- Uppercase letters (65–90) – A to Z
- Lowercase letters (97–122) – a to z
- Punctuation & symbols (e.g., !, @, #, \$)

3. Extended ASCII (128–255) (Non-Standard)

- Not part of original ASCII; varies by encoding (e.g., ISO-8859-1, Windows-1252).
- Includes special symbols (e.g., ©, ±, μ) and accented letters (e.g., é, ü).
- Key Points

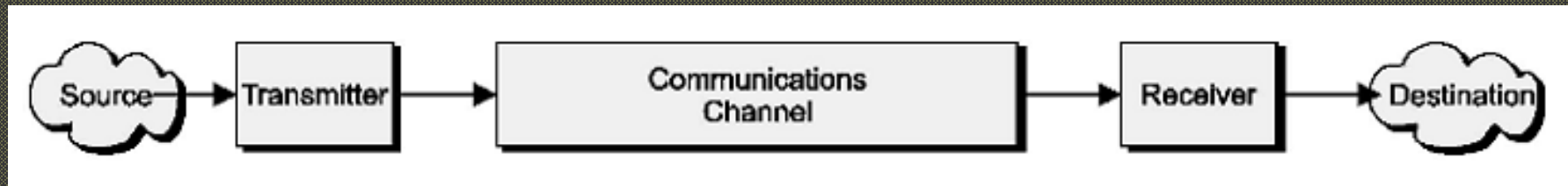
ASCII control characters			ASCII printable characters			Extended ASCII characters		
00	NULL	(Null character)	32	space	64	@	96	`
01	SOH	(Start of Header)	33	!	65	A	97	a
02	STX	(Start of Text)	34	"	66	B	98	b
03	ETX	(End of Text)	35	#	67	C	99	c
04	EOT	(End of Trans.)	36	\$	68	D	100	d
05	ENQ	(Enquiry)	37	%	69	E	101	e
06	ACK	(Acknowledgement)	38	&	70	F	102	f
07	BEL	(Bell)	39	'	71	G	103	g
08	BS	(Backspace)	40	(	72	H	104	h
09	HT	(Horizontal Tab)	41	)	73	I	105	i
10	LF	(Line feed)	42	*	74	J	106	j
11	VT	(Vertical Tab)	43	+	75	K	107	k
12	FF	(Form feed)	44	,	76	L	108	l
13	CR	(Carriage return)	45	-	77	M	109	m
14	SO	(Shift Out)	46	.	78	N	110	n
15	SI	(Shift In)	47	/	79	O	111	o
16	DLE	(Data link escape)	48	0	80	P	112	p
17	DC1	(Device control 1)	49	1	81	Q	113	q
18	DC2	(Device control 2)	50	2	82	R	114	r
19	DC3	(Device control 3)	51	3	83	S	115	s
20	DC4	(Device control 4)	52	4	84	T	116	t
21	NAK	(Negative acknowl.)	53	5	85	U	117	u
22	SYN	(Synchronous idle)	54	6	86	V	118	v
23	ETB	(End of trans. block)	55	7	87	W	119	w
24	CAN	(Cancel)	56	8	88	X	120	x
25	EM	(End of medium)	57	9	89	Y	121	y
26	SUB	(Substitute)	58	:	90	Z	122	z
27	ESC	(Escape)	59	;	91	[	123	{
28	FS	(File separator)	60	<	92	\	124	
29	GS	(Group separator)	61	=	93	]	125	}
30	RS	(Record separator)	62	>	94	^	126	~
31	US	(Unit separator)	63	?	95	_		
127	DEL	(Delete)						
128	Ç		160	á		192	Ł	
129	ü		161	í		193	ł	
130	é		162	ó		194	Ł	
131	â		163	ú		195	ł	
132	ä		164	ñ		196	—	
133	à		165	Ñ		197	Ł	
134	á		166	ª		198	ä	
135	ç		167	º		199	Ä	
136	ê		168	¿		200	Å	
137	ë		169	®		201	Ł	
138	è		170	¬		202	Ł	
139	ï		171	½		203	Ł	
140	î		172	¼		204	Ł	
141	í		173	¡		205	—	
142	Ä		174	«		206	Ł	
143	Å		175	»		207	Ł	
144	É		176	•		208	ø	
145	æ		177	•		209	Đ	
146	Æ		178	•		210	Ê	
147	ö		179	•		211	É	
148	ö		180	•		212	Ê	
149	ò		181	À		213	ı	
150	û		182	Á		214	ı	
151	ù		183	Â		215	ı	
152	ÿ		184	•		216	ı	
153	Ö		185	•		217	ı	
154	Ü		186	•		218	ı	
155	ø		187	•		219	ı	
156	£		188	•		220	ı	
157	Ø		189	•		221	ı	
158	×		190	•		222	ı	
159	f		191	ı		223	ı	
								nbsp

<https://theasciicode.com.ar/>

1.3 TRANSMITTERS, RECEIVERS, AND COMMUNICATION CHANNELS

For effective communication (e.g., data transmission), the following elements are essential:

- Source – The originator of the information (e.g., a person, sensor, or computer).
- Transmitter – Encodes the information into a signal suitable for the transmission medium (e.g., modulates digital data into radio waves, electrical pulses, or optical signals).
- Communication Channel – The physical or wireless medium carrying the signal (e.g., fiber-optic cable, air, or Ethernet).
- Receiver – Decodes the transmitted signal back into usable information (e.g., demodulates a radio signal into digital data).
- Destination – The intended endpoint that receives and interprets the information (e.g., a server, display device, or user).



Considerations

- Noise & Interference: The channel may introduce distortions, requiring error-checking mechanisms.
- Protocols: Standards (e.g., TCP/IP) ensure correct encoding, transmission, and decoding.
- Bidirectional Systems: Many systems (e.g., the Internet) use transceivers that act as both transmitters and receivers.

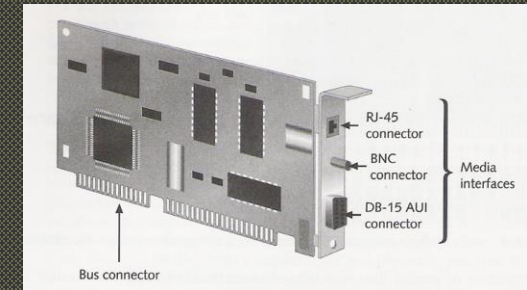
1.3.1 INTERFACE STANDARDS

- An interface standard (e.g., EIA/TIA-232, now commonly known as RS-232) defines the physical and electrical specifications required for communication between two or more systems or devices.
- These standards ensure compatibility but do not govern the protocol or method of data transfer (e.g., data encoding, framing, or error handling).

Characteristics Defined by Interface Standards

- Electrical Specifications
 - Voltage levels (e.g., $\pm 12\text{V}$ for RS-232 logic levels)
 - Signal timing (e.g., baud rate, rise/fall times)
 - Grounding and noise immunity requirements
- Mechanical Specifications
 - Connector type (e.g., DB-9, DB-25 for RS-232)
 - Pin assignments (e.g., Tx, Rx, RTS, CTS)
 - Cable length limitations
- Functional Specifications
 - Control signals (e.g., RTS/CTS for hardware flow control)
 - Timing sequences (e.g., handshaking procedures)
 - Signal definitions (e.g., DTR for "Data Terminal Ready")

RTS – Request to Send
CTS – Clear to Send



1.3.2 CODING

Coding (or character encoding) is the process of converting information (e.g., text, commands, or control signals) into a standardized digital format that can be transmitted and interpreted by communication devices. It ensures compatibility between systems by defining how data is represented as binary sequences.

1. ASCII (American Standard Code for Information Interchange)

Purpose: The most widely used encoding for text-based communication.

Bit Length: 7-bit (128 characters) or 8-bit (256 characters in extended ASCII).

Character Set:

Control characters (0–31 & 127): Non-printable (e.g., BEL, CR, LF).

Printable characters (32–126): Letters (A–Z, a–z), digits (0–9), and symbols (!, @, #, etc.).

Extended ASCII (128–255): System-dependent (e.g., accented letters, currency symbols).

Applications: Email, HTTP, plaintext files, and legacy systems.

2. EBCDIC (Extended Binary Coded Decimal Interchange Code)

Purpose: Primarily used in IBM mainframe and midrange systems.

Bit Length: 8-bit (256 characters).

Character Set:

Each character is assigned a unique Hexadecimal number, differing from ASCII's layout.

Includes letters, digits, and special symbols but lacks lowercase letters in early versions.

Applications: Banking, finance, and legacy enterprise systems.

3. Unicode (Modern Standard)

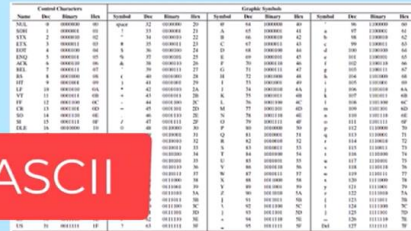
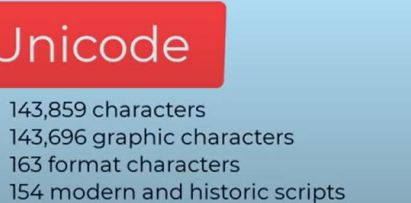
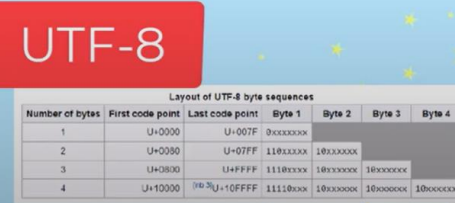
Purpose: Supports global scripts (e.g., Arabic, Chinese) and emojis.

Bit Length: Variable (UTF-8, UTF-16, UTF-32).

Advantages: Backward-compatible with ASCII (UTF-8 matches ASCII for 0–127).

Covers over 1 million characters.

Applications: Web pages (HTML), mobile apps, and international software.

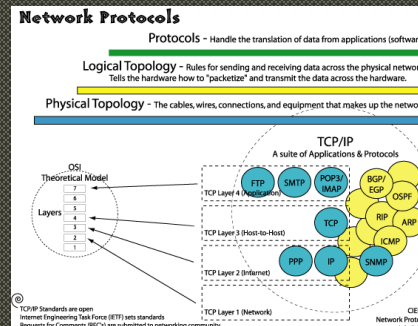
	
ASCII	EBCDIC
	
Unicode	UTF-8
143,859 characters 143,696 graphic characters 163 format characters 154 modern and historic scripts	

1.3.3 PROTOCOLS

Protocols (e.g., Modbus, Profibus, Data Highway Plus) are standardized sets of rules that govern how devices communicate over a network. They ensure reliable, efficient, and error-free data exchange by defining procedures for interaction, data formatting, and error handling.

Features of Communication Protocols

- **Connection Establishment & Initialization:** Handshake procedures (e.g., TCP's three-way handshake); Device authentication and parameter negotiation (e.g., baud rate in serial protocols).
- **Framing & Synchronization:** Start/End Delimiters - Mark boundaries of data frames (e.g., STX/ETX in ASCII-based protocols); Clock Synchronization - Asynchronous (e.g., UART uses start/stop bits); Synchronous (e.g., SPI relies on a shared clock signal).
- **Flow Control:** Prevents data overflow by regulating transmission speed.
- **Role Reversal:** Half/Full Duplex
- **Error Detection & Handling:** Parity bits (odd/even), Checksums (e.g., CRC in Modbus), Sequence numbers (e.g., TCP packets), Recovery - Retransmission (e.g., TCP) or abort (e.g., CAN bus error frames).
- **Timeout & Retry Mechanisms:** Defines response time limits (e.g., Modbus's 1.5-character timeout); Automatic retries or fallback actions (e.g., switching to backup servers).



- **STX – Start of Text: Decimal 2 – Binary 10**
- **ETX – End of Text: Decimal 3 – Binary 11**
- **UART - Universal Asynchronous Receiver/Transmitter**
- **SPI - Serial Peripheral Interface**

1.3.4 STANDARDS VS PROTOCOLS

1. Physical Layer Standards (The "Wiring Rules")

These define the electrical and mechanical characteristics of communication hardware but don't specify message content:

- RS-232: Serial communication standard
- RS-485: Robust serial standard
- Ethernet (IEEE 802.3): Network standard

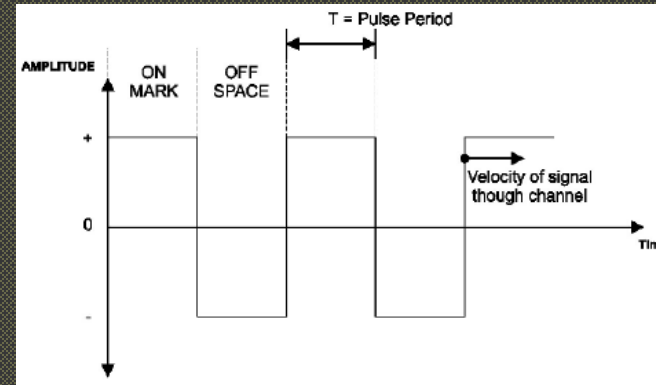
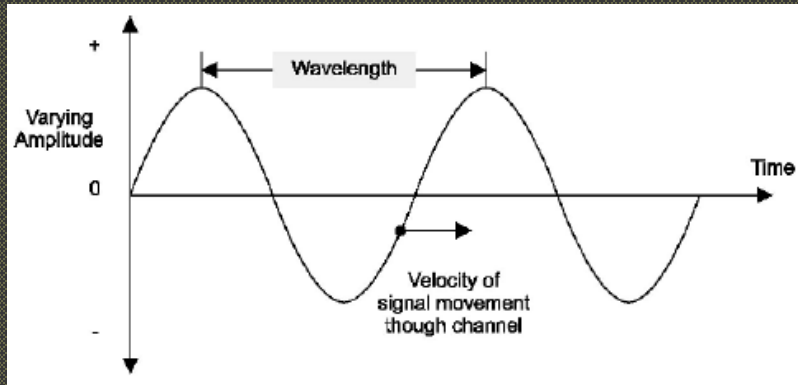
2. Communication Protocols (The "Language Rules")

These define how messages are structured and processed on top of physical standards:

- TCP/IP: Connection-oriented
- UDP/IP: Connectionless

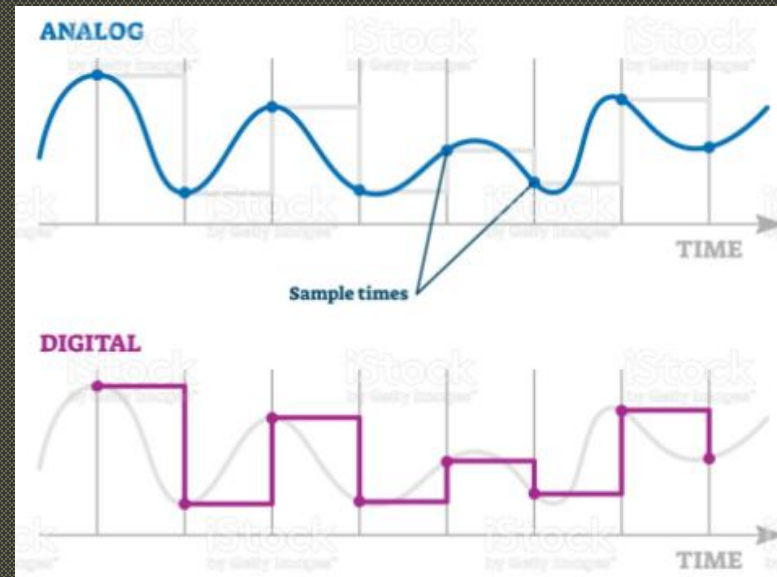
Protocol	Standard Used	Key Features	Common Applications
Modbus	RS-485/Ethernet	Simple master/slave, 16-bit registers	PLC communication
DH-485	RS-485	Allen-Bradley proprietary	Legacy AB PLC networks
DeviceNet	CAN bus	CIP adaptation, device profiles	Factory automation

1.4 ANALOG VS DIGITAL COMMUNICATION SIGNALS



1. Analog Signals

- A continuous signal representing physical measurements (e.g., 4-20mA Current Loops or 0-10V Voltage Signals).
 - Smooth, infinite values
- Pros:
 - Natural representation of real-world phenomena (e.g., temperature, pressure).
 - Higher theoretical resolution (infinite precision: e.g., 12.7mA = 54.2% flow rate).
- Cons:
 - Susceptible to EMI.
 - Hard to store/process (e.g., temperature records on paper).



2. Digital Signals

- Discrete values represented as binary numbers (e.g., "0"s and "1"s").
 - RS-485 (Modbus RTU), Ethernet/IP.
- Pros:
 - Noise immunity (error-checking via CRC).
 - Diagnostics (e.g., device health monitoring).
- Cons:
 - Higher setup complexity.
 - Requires protocol-specific hardware.

- Analog Dominates: Slow processes (temperature, level) still use 4-20mA for simplicity.
- Digital Rising: Ethernet-based protocols (EtherNet/IP, PROFINET) enable Industry 4.0.
- Hybrid Solutions: HART superimposes digital data on 4-20mA for diagnostics.

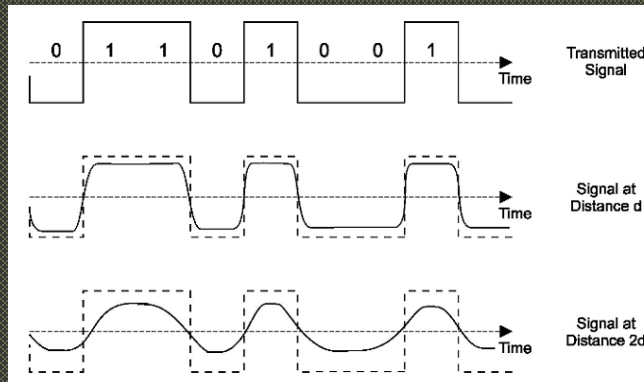
1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.1 SIGNAL ATTENUATION

- Attenuation means that a signal becomes weaker as it travels through a cable, air, or another communication medium.
- At the transmitter: The digital signal has clear HIGH and LOW levels, for example: 0 1 1 0 1 0 0 1
- At distance d: The signal is weaker, but the receiver can still clearly recognize the 0s and 1s.
- At distance 2d: The signal becomes even weaker and its edges may become less sharp.
- If the signal becomes too weak or distorted, the receiver may mistake a 0 for a 1, or a 1 for a 0: Causing data errors/corruption.
- Communication standards therefore specify maximum recommended cable lengths. For example:
 - Ethernet over Cat6: typically up to 100 m for 1 Gbps Ethernet.
 - RS-485: can reach ~ 1200 m at lower data rates, depending on cable quality, data rate, termination, and electrical noise.
- In general: Longer distance → weaker signal → greater chance of communication errors.

Several physical effects weaken the signal:

- Resistance: Copper wires resist electrical current. Some signal energy is converted into heat.
- Capacitance and inductance present in every cable can distort fast-changing signals, especially at high frequencies.
- Electromagnetic losses: Walls, objects, and even air can absorb, reflect, or scatter electromagnetic waves.



$$\text{Attenuation } (\alpha) \text{ in dB/m} = 10 \times \log_{10}(P_t/P_r)$$

Where:

- P_t = Transmitted power
- P_r = Received power
- At Distance 2d, if attenuation exceeds receiver sensitivity, the signal becomes unusable.

1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.1 SIGNAL ATTENUATION

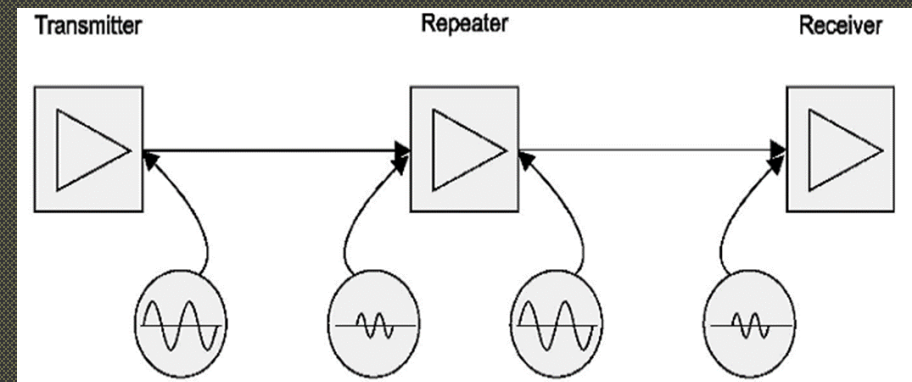
Ethernet (CAT6):

Frequency-Dependent Loss:

$$\alpha(f) = 20 \log_{10}(e) \times \sqrt{f} \times \left(\frac{R}{2Z_0} + \frac{GZ_0}{2} \right)$$

Where:

- $f = 250 \text{ MHz}$ (for 1Gbps Ethernet)
- $Z_0 = 100 \Omega$ (characteristic impedance)
- R, G = Resistance/conductance per unit length.

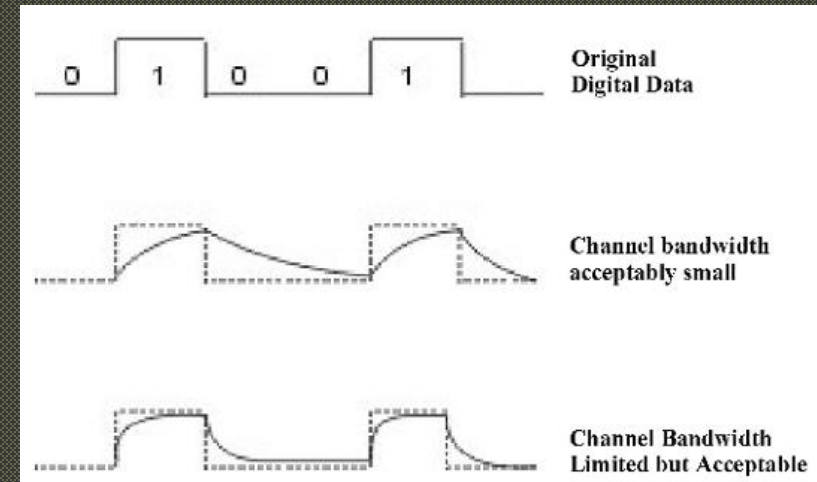


Method	How It Works	Example
Amplifiers/Repeaters	Boost signal power at intervals	Fiber-optic repeaters
Lower Resistance Cables	Use thicker/shorter conductors	AWG24 vs. AWG28 Ethernet cables
Equalization	Compensate for frequency-dependent loss	DSL modems
Protocol Resilience	Error-checking (CRC, retransmission)	TCP/IP, Modbus RTU

1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.2 CHANNEL BANDWIDTH

- Bandwidth describes the range of frequencies that a communication channel can carry successfully, measured in Hertz (Hz).
 - 1 kHz = 1,000 Hz
- A channel with higher bandwidth can carry faster-changing signals with less distortion. For example:
 - Cat6 cable has a bandwidth of ~250 MHz - high-frequency signal components needed for high-speed communication
- Think of a communication channel like a highway:
 - High bandwidth = a wider highway with more capacity.
 - Low bandwidth = a narrower highway with less capacity.
 - More bandwidth generally allows higher data rates.



- Digital signals contain rapid changes between HIGH and LOW levels.
- A square wave is actually made from several frequencies called harmonics. For example, a 1 kHz square wave contains:
 - 1 kHz - fundamental frequency; 3 kHz - 3rd harmonic; 5 kHz - 5th harmonic; 7 kHz - 7th; and higher frequencies
- These higher-frequency components help create the sharp edges of the digital signal.
- If the channel bandwidth is too low, some harmonics are removed.
- This can make it harder for the receiver to determine whether the signal represents a 0 or 1, causing communication errors.

1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.2 CHANNEL BANDWIDTH

- The data rate and bandwidth terms are related, but they are not the same.
 - Bandwidth: Measured in Hz; describes the frequency range the channel can carry.
 - Data Rate: Measured in bits per second (bps); describes how many bits are actually transmitted each second.
- Example: 100 Mbps = 100 million bits transmitted per second.
- Suppose we send a 1 kHz square wave through a channel with only 2 kHz bandwidth.
- The channel can pass the 1 kHz fundamental frequency, but it cannot properly pass the 3 kHz harmonic, 5, 7, or higher.
 - Therefore, the square wave becomes more rounded and distorted.
- A useful simplified rule is:
 - $\text{Bandwidth} \geq 3 \times \text{Fundamental Frequency}$
- For a 1 kHz square wave:
 - $\text{Bandwidth} \geq 3 \times 1\text{kHz} = 3\text{kHz}$
- Including even more harmonics gives a better approximation of the original square wave.
 - A simple way to remember it is:
 - Bandwidth (Hz) = capability of the channel
 - Data rate (bps) = amount of digital data being transmitted
 - Using the highway analogy:
 - Bandwidth = size of the highway
 - Data rate = number of cars passing each second
- Nyquist Theorem for ideal channel using two signal levels (0 and 1):
 - Maximum Data Rate $\approx 2 \times \text{Bandwidth}$
 - Therefore: $\text{Bandwidth} \approx \text{Data Rate} / 2$
- For a data rate of 9600 bps, the theoretical minimum bandwidth is:
 - $B = 9600 / 2 = 4800 \text{ Hz}$ or 4.8 kHz
- So, 9600 bps requires at least about 4.8 kHz of bandwidth.
- In real systems, additional bandwidth is normally needed because of noise, filtering, encoding, and cable characteristics.

1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.3 BAUD RATE VS DATA TRANSFER RATE

- Data Transfer Rate (bps):
 - Measures only the useful data transmitted
 - In the figure: 7 data bits are transmitted
 - Example: If this pattern repeats 100 times/second:
 - Data rate = 7 bits $\times$ 100 = 700 bps
- Baud Rate:
 - Counts all signal changes (including overhead)
 - In the figure: 10 total bits (7 data + 3 overhead)
 - Example: If this pattern repeats 100 times/second:
 - Baud = 10 bits $\times$ 100 = 1000 baud
- More stop bits = better noise immunity but lower efficiency
- Parity bits add reliability but reduce throughput

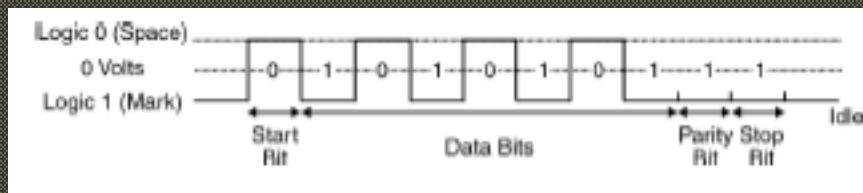
Overhead Calculation:

- Total bits: 10
- Data bits: 7
- Overhead: 3 bits (30% overhead)
 - 1 start bit
 - 1 parity bit
 - 1 stop bit

$$\text{Data Rate (bps)} = \frac{\text{Data Bits per Frame} \times \text{Baud Rate}}{\text{Total Bits per Frame}}$$

A PLC reads temperature data from a sensor using:

- Baud rate: 9,600 baud
- Frame format:
 - 1 start bit
 - 8 data bits
 - No parity (0 parity bits)
 - 1 stop bit
- Total bits per frame: 1 + 8 + 0 + 1 = 10 bits
- Each frame carries 8 data bits (the actual payload)
- Data Rate = $\frac{8 \text{ Data Bits} \times 9600 \text{ baud}}{10 \text{ total bits}} = 7,680 \text{ bps}$
- Theoretical max: 9,600 baud (signal changes/sec)
- Actual useful data rate: 7,680 bps (20% overhead from start/stop bits)



1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.4 NOISE

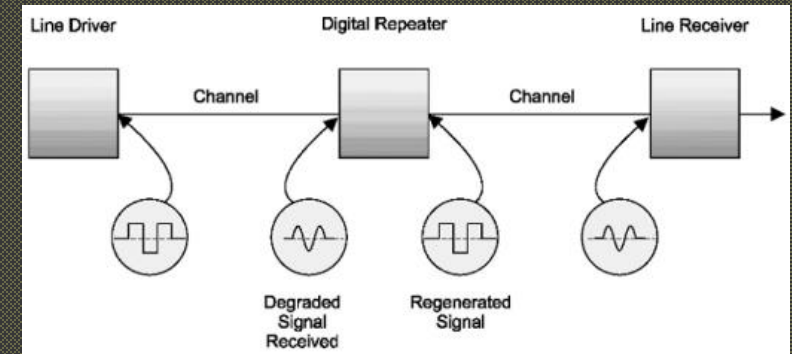
- Noise is unwanted interference that can distort communication signals.
- Common causes include thermal noise, EMI, crosstalk, motors, VFDs, and switching devices.
- Noise becomes more harmful when the signal is weakened by attenuation.

Signal-to-Noise Ratio (SNR)

- $SNR(dB) = 10 \log_{10} (S/N)$
- Higher SNR = cleaner signal
- Lower SNR = greater chance of communication errors
- There is no universal 10 dB limit; acceptable SNR depends on the system.

Example:

- Signal = 1 mW, Noise = 0.01 mW $\rightarrow$ SNR = 20 dB
- A PLC Ethernet cable installed near a VFD cable may experience EMI and communication errors.
 - Solution: Use proper cable separation, shielding, grounding, or fiber optic communication.



Analog vs. Digital

- Analog: Noise accumulates and is amplified with the signal.
- Digital: Repeaters/regenerators can re-amplify, reshape, and retime the signal.

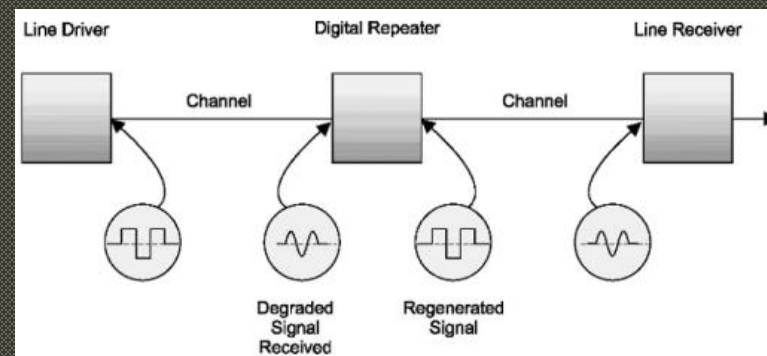
1.5 COMMUNICATION CHANNEL PROPERTIES

1.5.4 NOISE

How Digital Repeaters Work:

- Receive Degraded Signal:
 - Attenuated amplitude + noise added.
- Extract Digital Data:
 - Compare signal to threshold:
 - Voltage > Threshold → "1".
 - Voltage < Threshold → "0".
- Regenerate Clean Signal:
 - Transmit a new, noise-free copy of the original.

Feature	RS-232	RS-485	Ethernet
Max Distance	15m	1200m	100m (CAT6)
Topology	Point-to-point	Multi-drop	Star (switched)
Noise Immunity	Low (single-ended)	High (differential)	Very High (twisted pair)
Regeneration	Not supported	Repeaters	Switches
Typical SNR	20–30 dB	25–40 dB	20–35 dB



1.6 DATA TRANSMISSION MODES

1.6.1 DIRECTION OF SIGNAL FLOW (SIMPLEX)

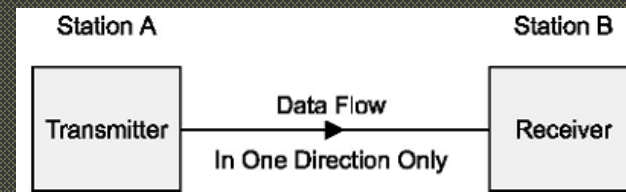
- Station A (Transmitter): Sends data unidirectionally (e.g., a sensor streaming readings).
- Station B (Receiver): Only receives data - no return path for acknowledgments or control signals.
- Data Flow: One direction only (→), like a one-way street.

- Critical Limitations:
 - No Process Control: Cannot send commands (e.g., "stop motor").
 - No Acknowledgment: Cannot confirm data receipt (risk of undetected errors).
 - No SNR Feedback: Cannot adapt transmission power to noise conditions.
- Exceptions:
 - Legacy Systems: Pager networks, some SCADA telemetry.
 - Broadcast-Only: Emergency alerts, public address systems.

Feature	Description
Directionality	Data flows only from transmitter to receiver.
Efficiency	100% bandwidth for one direction (no reverse path overhead).
Limitations	No error detection/retransmission, no flow control.

Real-World Examples

- Public Radio Broadcasting (FM/AM):
 - Transmitter → Radio receivers (no uplink).
 - SNR Consideration: Weak signals (e.g., rural areas) cause noise, but receivers cannot request retransmission.
- Industrial Applications:
 - Sensor-to-Display Units: Temperature sensors streaming to a non-interactive HMI.
 - Safety Alarms: Fire alarms triggering alerts without feedback.



1.6 DATA TRANSMISSION MODES

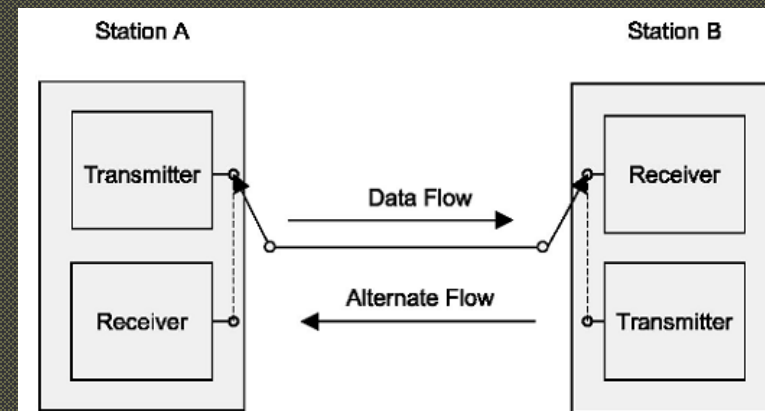
1.6.1 DIRECTION OF SIGNAL FLOW (HALF DUPLEX)

- Station A: Operates as Transmitter (sending data) or Receiver (receiving data), but not simultaneously.
- Station B: Similarly switches roles between transmitting and receiving.
- Data Flow: Alternates direction ($\leftrightarrow$) with a line turnaround procedure between transmissions.

- Station A Transmits: Sends data to Station B (e.g., PLC sending a command to a motor drive).
- Line Turnaround:
 - Station A stops transmitting and switches to receive mode.
 - Station B switches from receive to transmit mode.
 - Delay: Typically 1 - 2 character times (e.g., 1.04 ms for 9600 baud Modbus).
- Station B Responds: Sends acknowledgment or data back to Station A.

Feature	Description
Directionality	Bidirectional, but one direction at a time.
Channel Sharing	Single channel (wire/frequency) shared for both directions.
Efficiency	Lower than full-duplex due to turnaround delays.
Industrial Use	Modbus RTU, HART, CAN bus.

HART - Highway Addressable Remote Transducer



1.6 DATA TRANSMISSION MODES

1.6.1 DIRECTION OF SIGNAL FLOW (HALF DUPLEX)

Real-World Examples

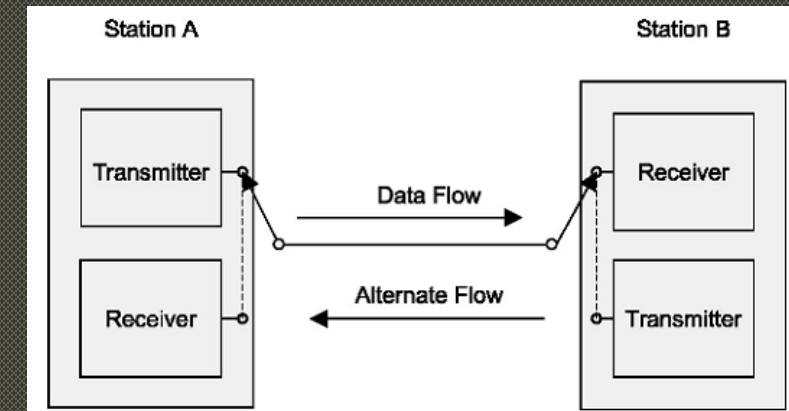
- Walkie-Talkies:
 - Users press "PTT" (Push-to-Talk) to switch from receive to transmit mode.
 - Industrial Equivalent: Two-way radios in factory settings.
- Modbus RTU (RS-485):
 - Master (PLC) sends a request → Slave (sensor) responds after turnaround.
 - Critical Detail: RS-485 transceivers use "DE/RE" pins to control direction.
- CAN Bus:
 - Devices arbitrate for bus access (priority-based), then release the bus for replies.

DE/RE - Driver Enable / Receiver Enable

- Cost Savings: Single pair of wires (RS-485) vs. dual pairs (full-duplex).
- Noise Resilience: Differential signaling (RS-485) cancels noise during turnaround.
- Determinism: Predictable delays suit slow/moderate-speed industrial networks.

Tradeoffs:

- Collision Risk: If two devices transmit simultaneously (e.g., misconfigured Modbus).
- Latency: Turnaround delays reduce throughput (e.g., 30% slower than full-duplex).

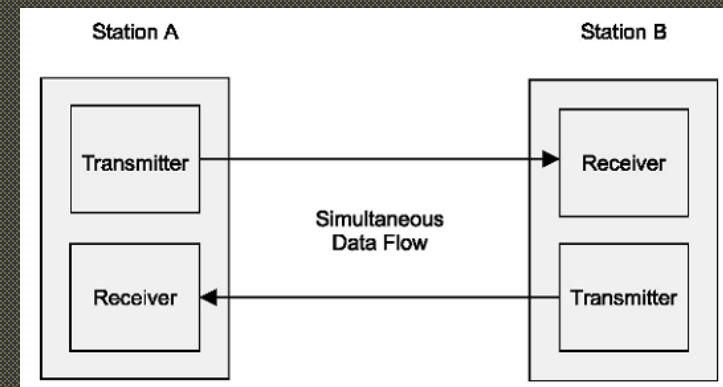


1.6 DATA TRANSMISSION MODES

1.6.1 DIRECTION OF SIGNAL FLOW (FULL DUPLEX)

- Station A: Simultaneously transmits and receives data (e.g., PLC sending commands while receiving sensor feedback).
- Station B: Likewise transmits and receives at the same time.
- Data Flow: Truly simultaneous ($\uparrow\downarrow$), like a two-way highway.
- Physical Layer:
 - Wired: Separate transmit/receive pairs (e.g., CAT5e Ethernet uses 4 wires: 2 pairs).
 - Wireless: Different frequencies for uplink/downlink (FDD in 4G/5G).
- Protocol Layer:
 - No turnaround delays (unlike half-duplex).
 - Collision avoidance via switches (Ethernet) or time-slicing (TDD in wireless).
- Industrial Example:
 - A motor controller receives speed commands while streaming real-time torque data to the PLC.

Feature	Description
Directionality	Simultaneous transmit/receive
Channel Usage	Dedicated paths for each direction (e.g., separate wires/frequencies)
Efficiency	100% bandwidth in both directions
Industrial Use	EtherNet/IP, PROFINET, RS-422



FDD - Frequency Division Duplexing
TDD - Time Division Duplexing

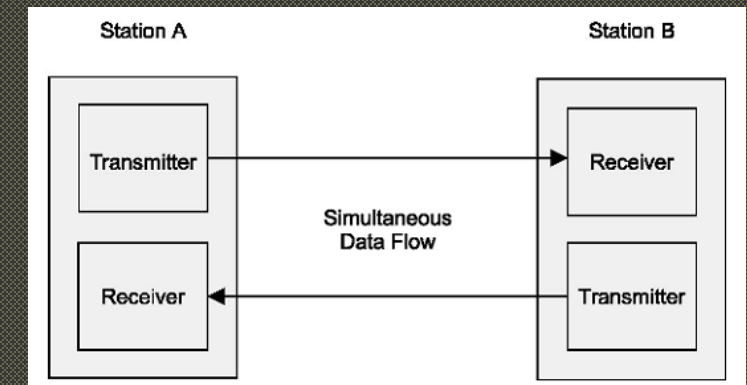
1.6 DATA TRANSMISSION MODES

1.6.1 DIRECTION OF SIGNAL FLOW (FULL DUPLEX)

Real-World Examples

- Industrial Ethernet (EtherNet/IP):
 - Full-duplex switched networks enable:
 - PLC ↔ HMI communication
 - I/O device polling without delays
 - RS-422:
 - Uses 4 wires (Tx+/Tx-, Rx+/Rx-) for noise-resistant full-duplex serial.
 - Zero Latency: Critical for real-time control (e.g., robot motion control).
 - Determinism: Enables precise cycle times (<1ms in PROFINET IRT).
 - Scalability: Switches allow star topologies with 100+ devices.
- Tradeoffs:
- Cost: More wiring/complexity than half-duplex.
 - Noise Management: Crosstalk in high-speed links (mitigated with shielding).

Aspect	Simplex	Half-Duplex	Full-Duplex
Direction	One-way	Alternating	Simultaneous
Industrial Use	Rare (alarms)	Modbus RTU	EtherNet/IP
Bandwidth	100% Tx	50% Tx/Rx	100% Tx + Rx

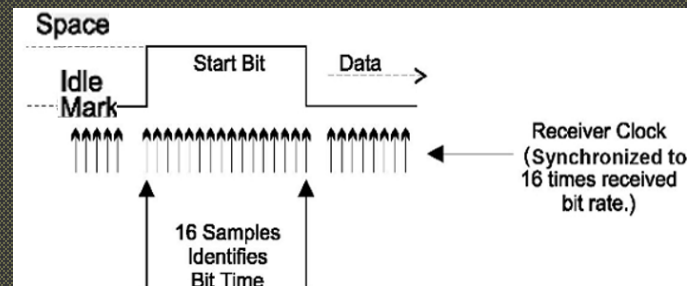


1.6 DATA TRANSMISSION MODES

1.6.2 SYNCHRONIZATION IN SERIAL COMMUNICATION

- The transmitter and receiver must agree on the bit rate (baud rate) so each bit is sampled at the correct time.
- This timing agreement is called synchronization.
- In UART, there is no separate clock signal.
- The receiver detects the start bit, then uses its own internal clock to sample the remaining bits.
- UART receivers often use oversampling (for example, 16×) to improve timing accuracy.
- Correct timing allows the receiver to distinguish between 0s and 1s and decode the data properly.

The receiver uses the start-bit transition to synchronize its internal clock, then samples each bit at the expected time.



1.6 DATA TRANSMISSION MODES

1.6.2 SYNCHRONIZATION IN SERIAL COMMUNICATIONS

- UART receivers often sample the signal at $16\times$ the baud rate.
- The falling edge of the start bit synchronizes the receiver.
- Data bits are then sampled near their center, where the signal is most stable.
- Oversampling improves tolerance to noise and small timing differences.

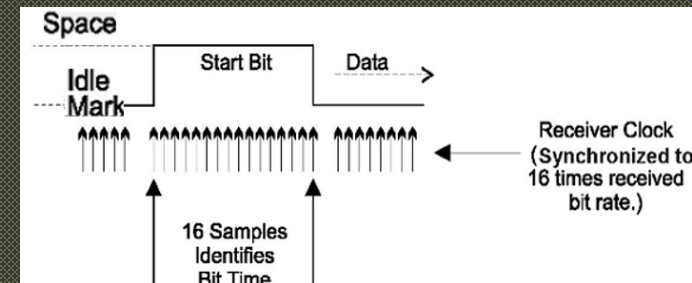
Example – 9600 bps

- Bit time $\approx 104\ \mu\text{s}$
- Sampling rate = 153.6 kHz
- Sample interval $\approx 6.5\ \mu\text{s}$

At 9600 bps, the transmitter sends 9600 bits every second.

- Bit Time $\approx 104\ \mu\text{s}$
 - The time for one bit is: Bit Time = $\frac{1}{9600} \approx 104.2\ \mu\text{s}$
- Sampling Rate = 153.6 kHz:
 - $9600 \times 16 = 153,600\ \text{samples/s}$
- Sample Interval $\approx 6.5\ \mu\text{s}$
 - The time between samples is: $\frac{104.2\ \mu\text{s}}{16} \approx 6.5\ \mu\text{s}$

Method	Pros	Cons	Used In
Oversampling	Simple, low-cost	Limited to medium speeds	UART, Modbus
PLLs	High-speed tolerant	Complex hardware	Ethernet, USB
Sync Bytes	Robust in noisy environments	Overhead (extra bytes)	Ethernet (Preamble and Start Frame Delimiter)



1.6 DATA TRANSMISSION MODES

1.6.2 SYNCHRONIZATION IN SERIAL COMMUNICATIONS

Sender

- Keeps the line HIGH while idle.
- Sends a LOW start bit to indicate the beginning of a frame.
- Sends the data bits.
- Ends the frame with a HIGH stop bit.

Receiver

- Detects the start-bit transition.
- Synchronizes its internal timing.
- Samples each data bit near its center.
- Reconstructs the transmitted data.

1. Idle State (Mark)

```
Sender:  _____ (Held high = Logic 1)
Receiver: "Listening for start bit..."
```

2. Start Bit (Space)

```
Sender:  _|^^|____ (Drops low for 1 bit time = "Start!")
          ↓
          Start Bit (Logic 0)
Receiver: "Falling edge detected! Timer starts NOW."
```

3. Data Transmission (Example: "1010")

```
Bit Time: |<--- 1 bit duration (e.g., 104µs @ 9600bps) --->|
Sender:   ___|^^|___|^^|___|^^|___|^^|___
          Bit1 Bit2 Bit3 Bit4
          "1"  "0"  "1"  "0"
```

4. Receiver Sampling (16x per Bit)

```
Sample points (16 per bit):
          ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
Bit:      |----- Bit 1 ("1") -----|
Samples:  1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1
          (Noise glitch at sample 9-10)
Decision: Majority vote → "1" (samples 8-10 = 1,1,0 → "1" wins)
```

5. Visualizing Oversampling

```
Ideal "1" Bit:  _____
Noisy "1" Bit:  _^^^_____
                1 1 0 1 1 1 0 1 1 1 1 0 1 1 1 1
                (Middle samples 8-10: ↑↑↑)
                Votes: 1,1,1 → "1" (rejects noise)
```

6. Stop Bit

```
Sender:   ___|^^|_____ (Rises high = Stop Bit)
Receiver: "Message complete!"
```

1.6 DATA TRANSMISSION MODES

1.6.3 ASYNCHRONOUS TRANSMISSION

- Sends data without a shared clock signal.
- Data is transmitted in separate frames.
- Each frame typically contains:
 - Start bit – indicates the beginning of the frame
 - Data bits – usually 8 bits
 - Optional parity bit – basic error checking
 - Stop bit(s) – indicate the end of the frame
- The communication line stays HIGH (Idle) between frames.
- Sender and receiver must use approximately the same baud rate.

Advantages

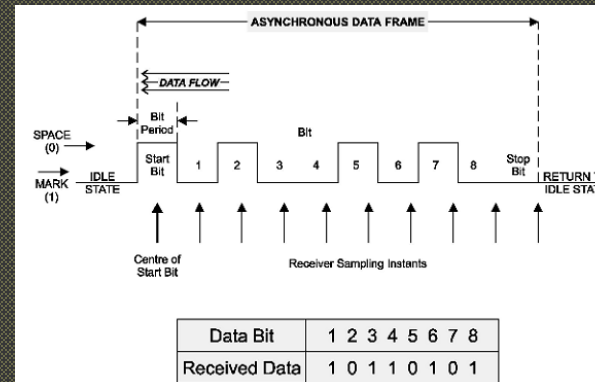
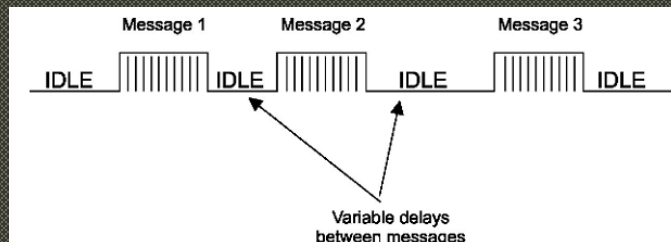
- Simple and inexpensive
- No separate clock wire required
- Allows variable delays between messages

Disadvantages

- Extra start/stop bits create communication overhead
- Less efficient than synchronous transmission
- UART parity provides only basic error checking

Common Applications

- RS-232 / RS-485 serial communication
- Microcontrollers and embedded devices
- GPS and barcode readers
- Modbus RTU



1.6 DATA TRANSMISSION MODES

1.6.4 SYNCHRONOUS TRANSMISSION

- Transmitter and receiver operate with coordinated timing.
- Timing may come from:
 - A shared clock, or
 - Clock information recovered from the signal.
- Data is usually sent as continuous blocks or frames.
- Unlike asynchronous transmission, individual characters do not require separate start and stop bits.
- This reduces overhead and supports higher-speed communication.
- Frames may include:
 - Synchronization information
 - Address/control information
 - Data
 - Error checking such as CRC/FCS

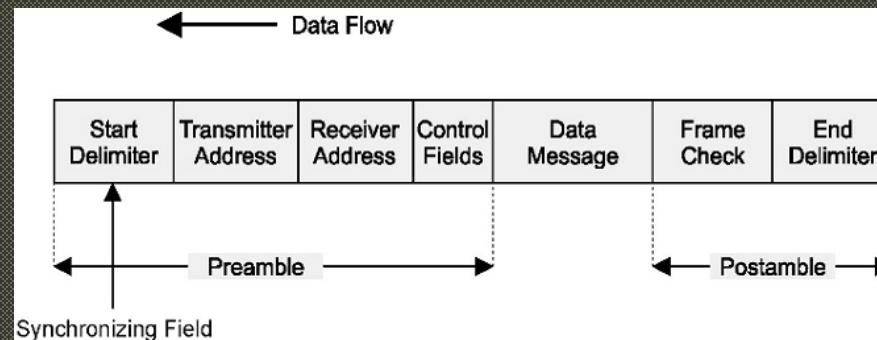
Examples

- SPI / I²C - use a separate clock signal.
- Ethernet - receiver recovers timing from the transmitted signal.

SPI and I²C are two common synchronous serial communication protocols used to connect microcontrollers with sensors, displays, memory chips, and other electronic devices.

SPI - Serial Peripheral Interface

I²C - Inter-Integrated Circuit



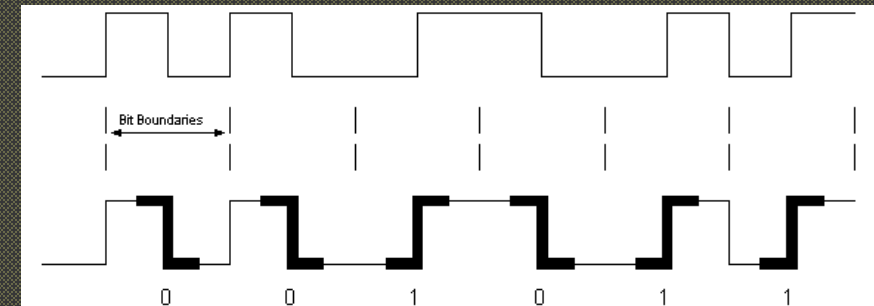
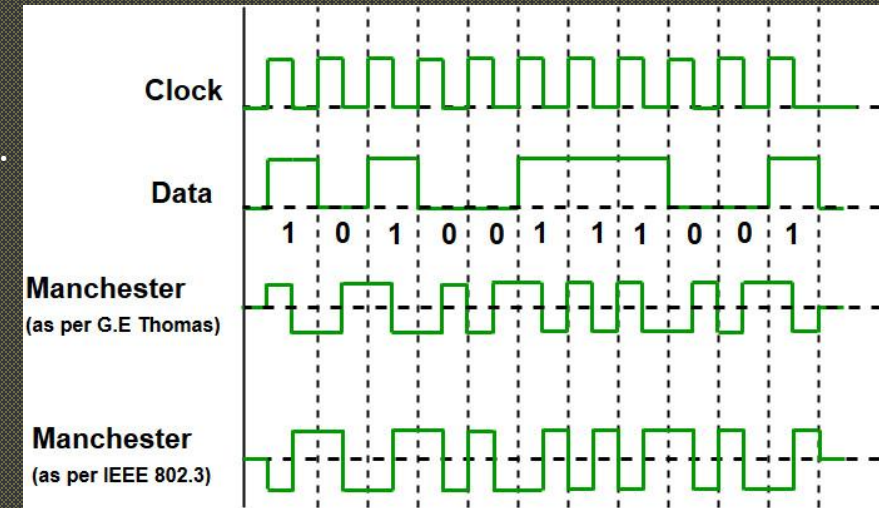
1.7 ENCODING METHODS

1.7.1 MANCHESTER

- Manchester encoding combines data and timing information in the same signal.
- Every bit has a transition in the middle, allowing the receiver to recover the clock.
- No separate clock wire is required.
- It prevents long periods without transitions.
- It is DC balanced, making it suitable for transformer-coupled communication.
- It requires more bandwidth than NRZ.

Aspect	IEEE 802.3	G.E. Thomas
0 Encoding	High→Low (↓)	Low→High (↑)
1 Encoding	Low→High (↑)	High→Low (↓)

Feature	Description
Self-Clocking	Embedded transitions prevent clock drift (even for 000.../111... sequences)
DC Balance	Equal high/low durations eliminate DC bias
Noise Immunity	Transitions are easy to detect even with noise



1.7 ENCODING METHODS

1.7.1 MANCHESTER

Applications

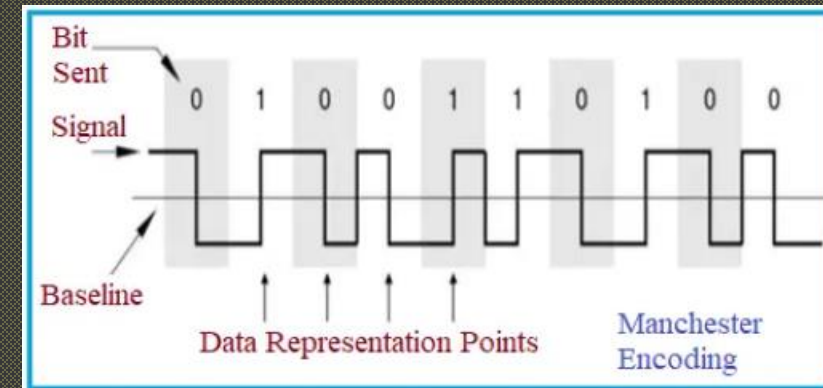
- 10 Mb/s Ethernet
- Some RFID systems
- PROFIBUS PA / MBP

Advantages

- Self-clocking
- No separate clock line
- Good DC balance
- Reliable clock synchronization

Disadvantages

- Higher bandwidth requirement
- More encoding/decoding than simple NRZ



<https://www.rfwireless-world.com/terminology/manchester-vs-differential-manchester-encoding>

1.7 METHODS

1.7.2 DIFFERENTIAL MANCHESTER

- Differential Manchester is a self-clocking line encoding method.
- Every bit contains a mid-bit transition for clock synchronization.
- Data is determined by whether a transition occurs at the start of the bit:
 - 0 → Transition at start
 - 1 → No transition at start
- The signal depends on changes in level, not the actual HIGH or LOW voltage.

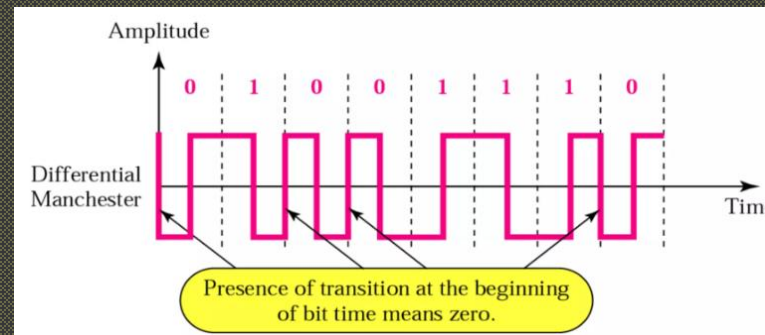
Historically used in IEEE 802.5 Token Ring networks

Advantages

- Self-clocking
- Good synchronization
- Polarity independent
- No separate clock line required

Disadvantages

- Requires more bandwidth than NRZ
- More complex than simple line encoding



1.7 ENCODING METHODS

1.7.3 RETURN TO ZERO

- RZ is a line-encoding method in which the signal returns to zero within each bit interval.
- This creates more transitions than NRZ, which can help with clock synchronization.
- It requires more bandwidth than NRZ.

Advantages

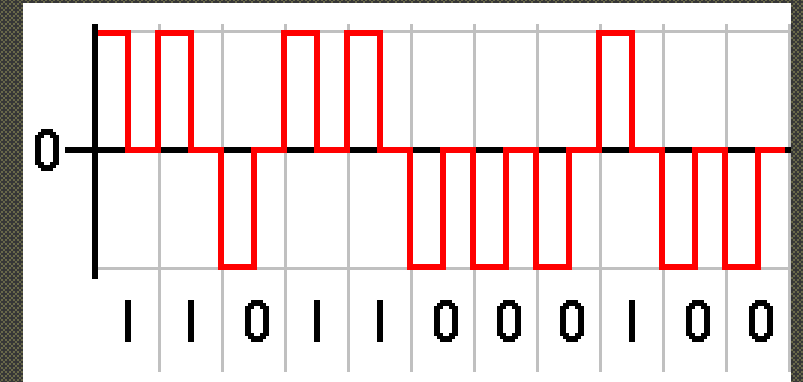
- Better timing synchronization
- Clear separation between pulses
- Useful in some optical communication systems

Disadvantages

- Higher bandwidth requirement
- Less bandwidth-efficient than NRZ
- Some RZ forms may still have synchronization problems with long sequences of zeros

Applications

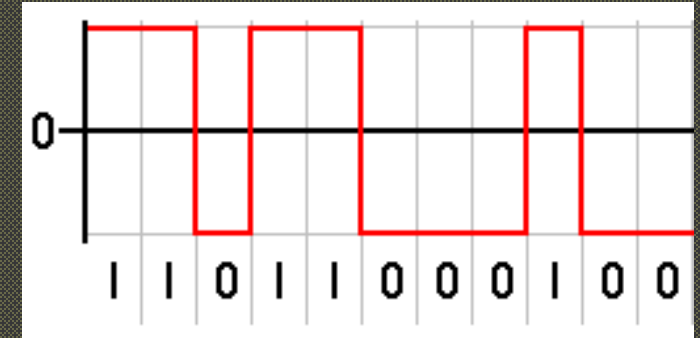
- Early telecommunications
- Some optical/fiber communication systems
- Legacy digital systems



1.7 ENCODING METHODS

1.7.4 NON-RETURN TO ZERO

- NRZ represents data using constant signal levels during each bit.
- In NRZ-L:
 - One voltage level represents 1
 - Another voltage level represents 0
- The signal does not return to zero between bits.
- NRZ uses less bandwidth than Manchester or RZ.
- Long sequences of identical bits produce no transitions, which can make synchronization difficult.



Advantages

- Simple encoding
- Good bandwidth efficiency
- Fewer signal transitions

Disadvantages

- Poor clock synchronization with long runs of 0s or 1s
- May require additional clock recovery, scrambling, or synchronization methods

Applications

- RS-232-style serial signaling
- Some optical communication systems
- Used as the basis for several digital encoding schemes

Encoding	Signal Behavior	Synchronization	Bandwidth
RZ	Returns to zero within each bit	Generally better timing information	Higher
NRZ	Maintains level for entire bit	Difficult with long identical-bit sequences	Lower
Manchester	Mid-bit transition every bit	Self-clocking	Higher

1.8 ERROR DETECTION AND CORRECTION

Sources of Communication Errors

Industrial networks may be affected by:

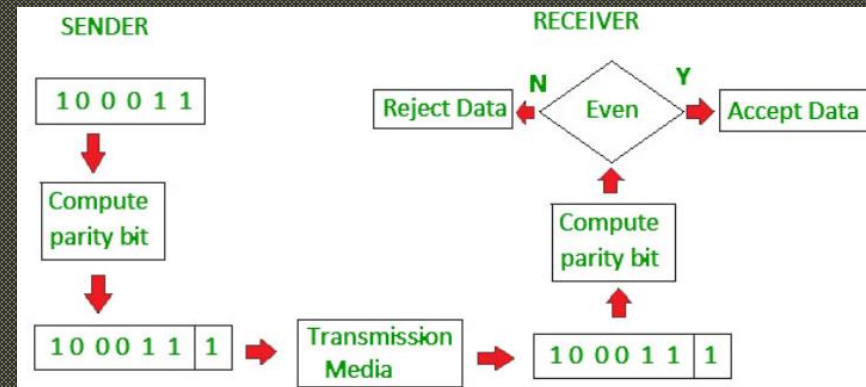
- VFD and motor noise
- Relay/contact switching
- Cable crosstalk
- Radio-frequency interference (RFI)

These may cause:

- Bit errors
- Corrupted messages
- Lost communication

Reducing Communication Errors

- Use shielded twisted-pair cable
- Separate communication cables from power cables
- Use proper grounding, bonding, and termination
- Use fiber optics in high-EMI environments
- Use repeaters when needed for long networks



<https://www.geeksforgeeks.org/computer-networks/error-detection-in-computer-networks/>

Error Detection

Communication protocols may use:

- Parity
- Checksum
- CRC

1.8 ERROR DETECTION AND CORRECTION

1.8.1 PARITY CHECK

- A parity bit is a single error-detection bit added to 7/8-bit data messages in asynchronous systems (like RS-232).
- It helps detect single-bit errors caused by noise.
- Sender: Calculates and appends the parity bit.
- Receiver: Recalculates parity and compares with received bit.
- Error Flag: Mismatch = 1-bit error detected (but not corrected).

Limitation:

- It cannot detect even-numbered bit errors (e.g., 2 flipped bits).
- No error correction capability.

Parity Type	Rule	Example (Data: 1011001)
Even Parity	Total 1s (data + parity) must be even	1011001 has 4 1s → Parity = 0 (keeps count even)
Odd Parity	Total 1s (data + parity) must be odd	1011001 has 4 1s → Parity = 1 (makes count odd)

1.8 ERROR DETECTION AND CORRECTION

1.8.1 PARITY CHECK

- Even parity - Even parity means the number of 1's in the given word including the parity bit should be even (2,4,6,...). The parity bit is set to 1 or 0 such that the no. of "1 bits" in the entire word is even.

P	Data Bits							
0	1	0	0	0	1	0	1	1

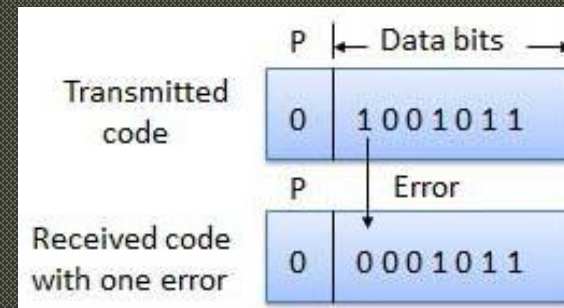
P	Data Bits							
1	0	0	0	0	1	0	1	1

- Odd parity - Odd parity means the number of 1's in the given word including the parity bit should be odd (1,3,5,...). The parity bit is set to 1 or 0 such that the no. of "1 bits" in the entire word is odd.

P	Data Bits							
1	1	0	0	0	1	0	1	1

P	Data Bits							
0	0	0	0	0	1	0	1	1

Application	Parity Type	Why It's Used
UART (RS-232)	Configurable	Simple noise detection for text/data
Industrial sensors	Odd Parity	Adds redundancy for noisy environments



1.8 ERROR DETECTION AND CORRECTION

1.8.2 CHECKSUM CALCULATIONS

Checksums (Simpler Alternative to CRC)

- Lightweight error detection for low-latency systems.
- Sum all data bytes (e.g., 8-bit or 16-bit).
- Append the least significant bits as checksum.
- Example: Data: 0x12, 0x34, 0x56 → Sum: $0x12 + 0x34 + 0x56 = 0x9C$ → Checksum: 0x9C

Technique	How It Works	Industrial Use Case
Parity Bits	Adds 1 extra bit to make 1s count even/odd	Simple sensors (temperature probes)
Checksums	Sums all data bytes; compares sent/received values	Modbus RTU (CRC-16 for basic checks)
Cyclic Redundancy Checks (CRC)	Polynomial division of data (e.g., CRC-32 detects >99.99% errors)	PROFINET, EtherCAT
Hamming Codes	Embeds error-correction bits to fix single-bit errors	Critical motor control systems
Forward Error Correction (FEC)	Transmits redundant data to reconstruct corrupted bits	Wireless HART in refinery networks

1.8 ERROR DETECTION AND CORRECTION

1.8.3 CYCLIC REDUNDANCY CHECK

Cyclic Redundancy Check (CRC)

Purpose: Detect *burst errors* (multi-bit errors) in high-speed networks.

How It Works:

- Polynomial Division:
 - Data bits are treated as a binary polynomial (e.g., $1101 = x^3 + x^2 + 1$)
 - Divided by a predefined generator polynomial (e.g., CRC -16: $x^{16} + x^{15} + 1$)
 - Remainder = CRC: Appended to the data frame.
 - Receiver Verification: Repeats division; non-zero remainder = error.
 - Detection: Catches 100% of errors $\leq$ polynomial degree (e.g., CRC-32 detects all ≤ 32 -bit errors).
 - Used In: Ethernet (CRC-32), USB, 5G.

Error Type	Why CRC-32 Catches It	Example
Single-Bit Errors	Any single-bit flip creates a non-zero remainder when divided by the prime polynomial.	1010 $\rightarrow$ 1011 (flipped LSB)
Burst Errors $\leq 32b$	The generator's 33-bit length ensures any burst ≤ 32 bits cannot be a multiple of the polynomial.	000...111...000 (≤ 32 flipped bits)
Odd # of Errors	CRC-32 detects all odd-bit errors (due to polynomial's $x + 1$ factor).	1, 3, 5, ..., 31-bit errors
Most Even-Bit Errors	Polynomial's irreducible nature catches most even-bit errors ($>99.99\%$).	2, 4, ..., 32-bit errors

1.8 ERROR DETECTION AND CORRECTION

1.8.3 CYCLIC REDUNDANCY CHECK

In CRC calculations, zero-padding ensures the data polynomial can be properly divided by the generator polynomial. The number of zeros added equals the degree of the generator polynomial (e.g., 10011 has degree 4, so append 4 zeros).

- Example

- Given Data: 1101011011
- Generator: 10011 (CRC-4, degree = 4)
- Append 4 zeros (degree of generator) to the data: →
- Treat the zero-padded data as a polynomial: → $1 \cdot x^{13} + 1 \cdot x^{12} + 0 \cdot x^{11} + \dots + 0 \cdot x^0$
- Divide by the generator polynomial 10011 ($x^4 + x + 1$) using binary division (XOR): →
- Append CRC to Data.

Original Data:	1 1 0 1 0 1 1 0 1 1
Zero-Padded:	1 1 0 1 0 1 1 0 1 1 0 0 0 0

$1 \cdot x^{13} + 1 \cdot x^{12} + 0 \cdot x^{11} + \dots + 0 \cdot x^0$

1101011011 0000 (Dividend)
XOR 10011 (Generator)

Final Remainder: 1110

- Transmitted frame: Original data + CRC remainder: 1101011011 1110

Feature	Parity Bit	CRC/Checksum
Error Detection	Single-bit only	Multi-bit bursts
Overhead	1 bit per 7-8 bits	16-32 bits per frame
Complexity	Simple (XOR gate)	Polynomial division

- Zero-padding ensures Full Division: The zeros reserve space for the remainder (CRC bits).
 - Aligns Polynomial Degrees: Allows the generator to "slide" over the entire data.
- Zero-Pad Length = Generator degree (n bits → append n zeros).
- Final CRC Length = n bits (same as generator degree).

1.8 ERROR DETECTION AND CORRECTION

1.8.4 HAMMING TECHNIQUE

Hamming Codes: Correct single-bit errors and fix them automatically (no need to ask for retransmission).

- Let's encode 1011 using the classic Hamming (7,4) code:
 - Insert Parity Bits: Positions: 1, 2, 4 (power-of-2 spots). P1, P2, P4.
 - Original data (1011) goes in positions 3,5,6,7:

Position:	1	2	3	4	5	6	7
	P1	P2	1	P4	0	1	1

Calculate Parity Bits

- P1 (Position 1 - 00**1**): Covers bits 1,3,5,7 → XOR of ?, 1, 0, 1 = 0 (to make even 1s). 000 00**1** 010 01**1** 100 10**1** 110 11**1**
- P2 (Position 2 - 0**1**0): Covers bits 2,3,6,7 → XOR of ?, 1, 1, 1 = 1.
- P4 (Position 4 - **1**00): Covers bits 4,5,6,7 → XOR of ?, 0, 1, 1 = 0.

Parity Bit	Binary Coverage Rule	Covers Positions (Example for Hamming 7,4)
P1 (1)	Positions where bit 0 (LSB) = 1; 000 00 1 010 01 1 100 10 1 110 11 1	1, 3, 5, 7 (binary: 1, 11, 101, 111)
P2 (2)	Positions where bit 1 = 1 000 001 0 1 0 0 1 1 100 101 1 10 1 11	2, 3, 6, 7 (binary: 10, 11, 110, 111)
P4 (4)	Positions where bit 2 = 1 000 001 010 011 1 00 1 01 1 10 1 11	4, 5, 6, 7 (binary: 100, 101, 110, 111)

Final encoded message: 0 1 1 0 0 1 1

Position: 1 2 3 4 5 6 7

1.8 ERROR DETECTION AND CORRECTION

1.8.4 HAMMING TECHNIQUE

Syndrome Calculation

- It is a 3-bit code (for Hamming 7,4) that acts like a treasure map to the error.
 - 000 = No errors.
 - Any other combo points to the bad bit.
- Example: We sent 0110011 but received 0110001 (bit 6 flipped).
 - Build the Syndrome
 - Write the results as P4 P2 P1:
 - $\rightarrow 1\ 1\ 0$ (binary for "6")
 - Fix the Error
 - Syndrome 110 = 6 $\rightarrow$ Flip bit 6 to fix:
 - 0110001 $\rightarrow$ 01100**1**1 (corrected!)

Check	Covers Positions	Calculation (XOR)	Result
P1	1,3,5,7	$0 \oplus 1 \oplus 0 \oplus 1 = 0$	✓
P2	2,3,6,7	$1 \oplus 1 \oplus 0 \oplus 1 = 1$	✗
P4	4,5,6,7	$0 \oplus 0 \oplus 0 \oplus 1 = 1$	✗

- Single-Bit Correction: Fixes errors like noisy Wi-Fi flipping a bit.
- Double-Bit Detection: Can detect (but not fix) 2-bit errors.
- Efficiency: Adds only 3 parity bits for 4 data bits (7 total).
- Limitation: Cannot correct 2-bit errors (use Reed-Solomon instead).

1.8 ERROR DETECTION AND CORRECTION

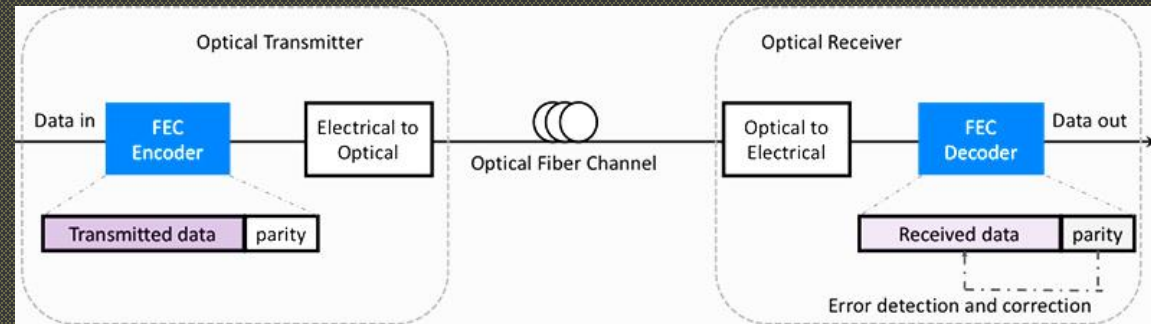
1.8.5 FORWARD ERROR CORRECTION

Forward Error Correction (FEC)

- When data is transmitted (e.g., over Wi-Fi or in space), noise can flip bits (0→1 or 1→0).
- Unlike error detection (which just flags errors), FEC fixes errors automatically without asking for retransmission.

Types of FEC

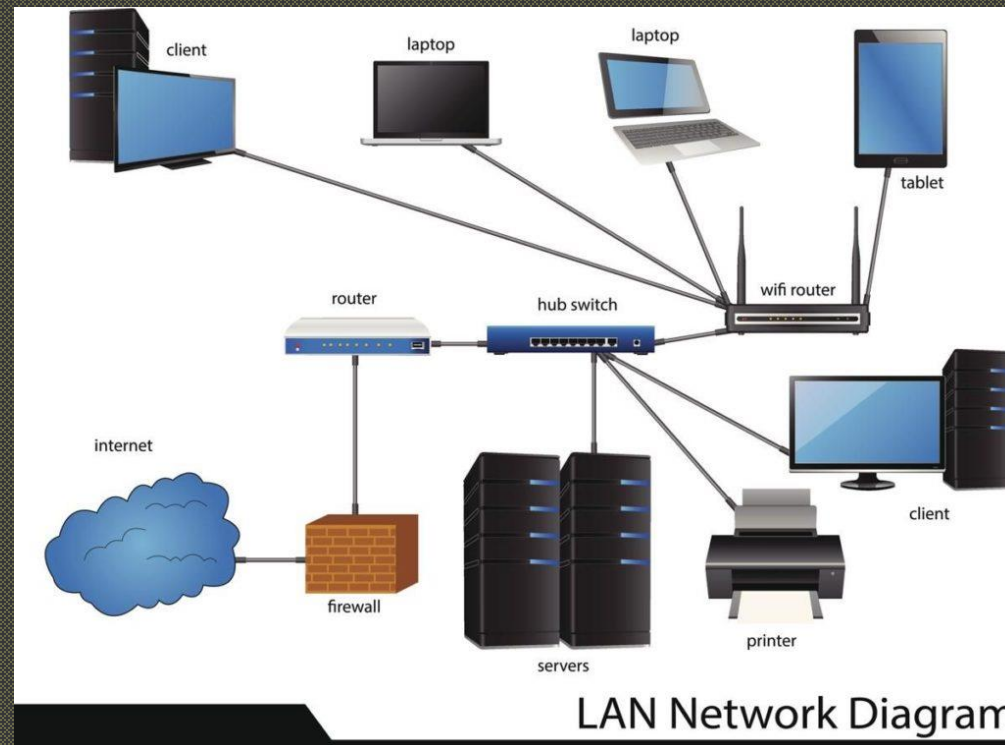
- Block Codes (e.g., Hamming, Reed-Solomon):
 - Work on fixed-size blocks (e.g., 4 data bits + 3 parity bits).
 - Used in QR codes, CDs/DVDs (Reed-Solomon).
- Convolutional Codes (e.g., Viterbi):
 - Work on continuous streams (no fixed blocks).
 - Used in Wi-Fi, satellite TV.
- Modern Codes (Polar Codes, LDPC):
 - Near-theoretical limits (used in 5G).



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

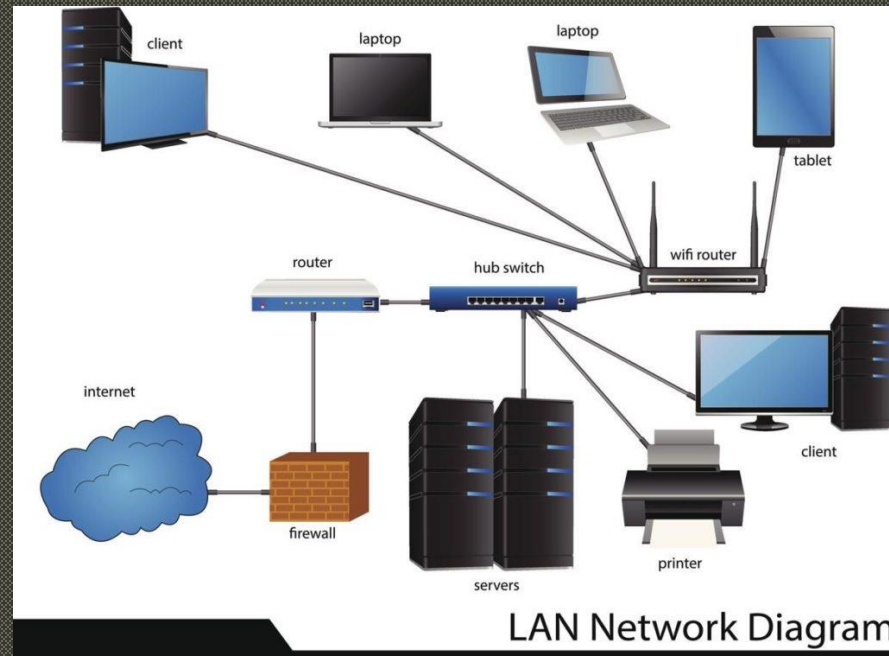
- A Local Area Network (LAN) is a high-speed communication network designed to interconnect devices within a limited geographical area, such as a single room, a factory floor, or an entire building.
- In industrial automation, LANs play a critical role in enabling real-time data exchange between computers, Programmable Logic Controllers (PLCs), Human-Machine Interfaces (HMIs), Industrial PCs (IPCs), sensors, and other smart devices.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

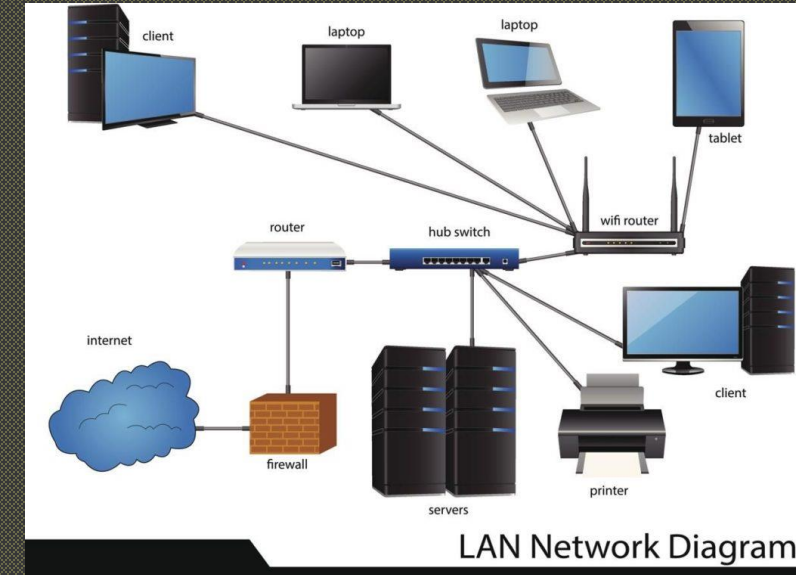
- Internet:
 - Represented by a cloud icon, indicating the external network connection point.
 - Serves as the gateway for the LAN to access external resources.
- Firewall:
 - Depicted as a brick wall icon between the internet and the router.
 - Acts as a security barrier, controlling incoming and outgoing traffic to protect the network from unauthorized access.
- Router:
 - Connects the LAN to the internet via the firewall and routes data between the network and external sources.
 - Also facilitates wireless connectivity (via Wi-Fi) for devices like laptops and tablets.
- Hub/Switch:
 - Functions as a central point for connecting wired devices within the LAN, distributing data packets to the appropriate destinations.
 - In modern networks, a switch is typically used, offering more efficient data handling than a traditional hub.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

- **Wi-Fi Router:**
 - Depicted with antenna-like extensions, connected to the hub/switch.
 - Provides wireless access points for devices, extending the network's reach to Wi-Fi-enabled equipment.
- **Servers:**
 - Represented by two stacked server towers.
 - Host data, applications, or services (e.g., file storage, email, or web hosting) accessible to devices on the LAN.
- **Clients:**
 - Include a desktop monitor (labeled "client") connected to the hub/switch and another monitor connected to a server.
 - Represent end-user devices (e.g., workstations or PCs) that access network resources and services.
- **Laptops:**
 - Shown as multiple laptop icons connected to the hub/switch and Wi-Fi router.
 - Portable devices that can connect via wired Ethernet or wirelessly, used for various tasks within the network.
- **Tablet:**
 - Illustrated as a tablet device connected to the Wi-Fi router.
 - A mobile device accessing the network wirelessly, typically for lightweight tasks or remote access.
- **Printer:**
 - A shared peripheral device that network users can access to print documents.



<https://www.shutterstock.com/image-vector/lan-network-diagram-icons-vector-illustrator-448271902?>

- **Wired Connections:** Lines that connect the router, hub/switch, servers, clients, laptops, and printer, indicating Ethernet or direct cable connections.
- **Wireless Connections:** Lines from the Wi-Fi router to laptops and the tablet signify wireless (Wi-Fi) connectivity.
- **Hierarchy:** The internet connects to the firewall, which feeds into the router. The router and hub/switch serve as central hubs, distributing connections to all devices, including the Wi-Fi router for wireless access.

1.9 TYPES OF NETWORKS

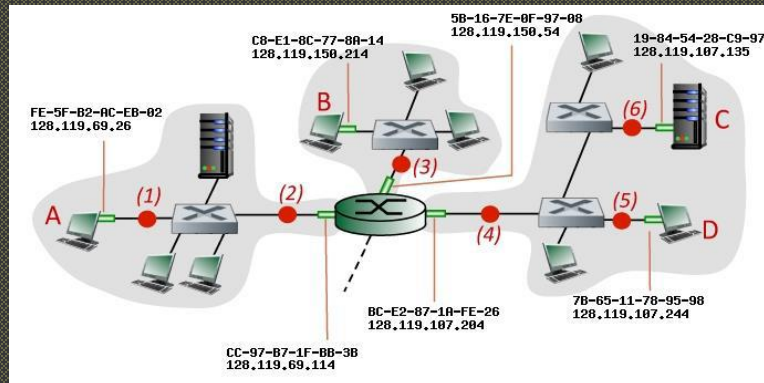
1.9.1 LOCAL AREA NETWORK (LAN)

Cabling Standards:

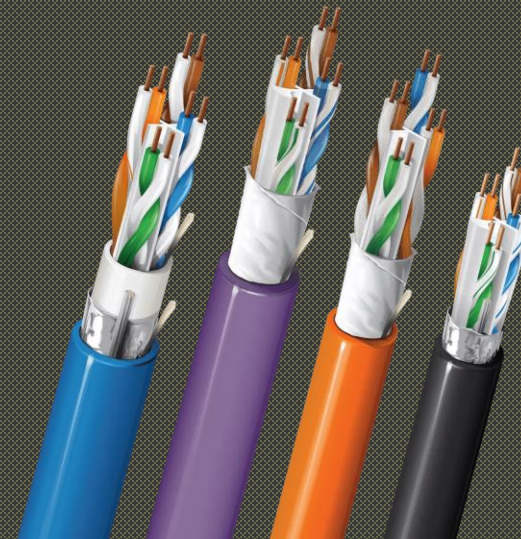
- Twisted Pair (CAT 6 / CAT 6A): Used for connecting end devices (PLCs, HMIs, PCs) to switches. CAT 6A supports 10 Gbps speeds, making it future-proof for industrial automation.
- Fiber Optic Cables: Used for backbone connections between switches, especially in environments with electromagnetic interference (EMI) or long-distance requirements (e.g., connecting different floors in a factory).

Nodes and Addressing:

- Each device connects to the LAN via a node/interface, which has a unique MAC/IP address.
- Every transmitted message includes the destination node's address, ensuring precise data delivery.



<https://www.chegg.com/homework-help/questions-and-answers/consider-figure-ip-mac-addresses-shown-nodes-b-c-d-well-router-s-interfaces-consider-ip-d-q25140485>

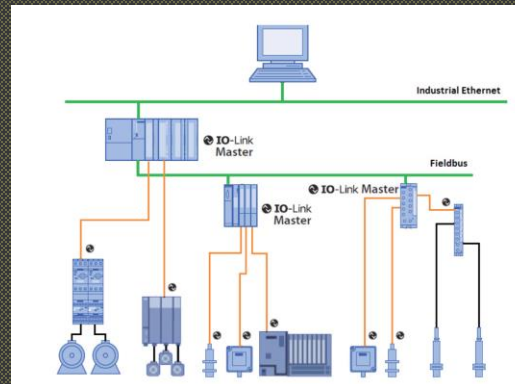


<https://www.belden.com/products/cable/ethernet-cable/category-6a-cable>

1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

- Factory Automation:
 - A PLC controlling a robotic assembly line sends real-time sensor data to an HMI and a central SCADA system over a LAN.
 - CAT 6A cables connect the PLC to an industrial switch, while fiber optics link switches across different production zones.
- Process Control in Oil & Gas:
 - Distributed Control Systems (DCS) rely on LANs to synchronize operations between remote I/O modules, flow meters, and control valves.
 - 10Base5 (Thick Ethernet) was historically used (10 Mbps over 500m), but modern plants now use Gigabit Ethernet (1000Base-T) for higher bandwidth.
- Legacy Systems & Signal Boosting:
 - In older installations, 10Base5 (Thick Ethernet) required repeaters to extend beyond 500m.
 - Today, fiber-optic media converters or industrial-grade Ethernet extenders are used to maintain signal integrity over long distances in harsh environments.



1.9 TYPES OF NETWORKS

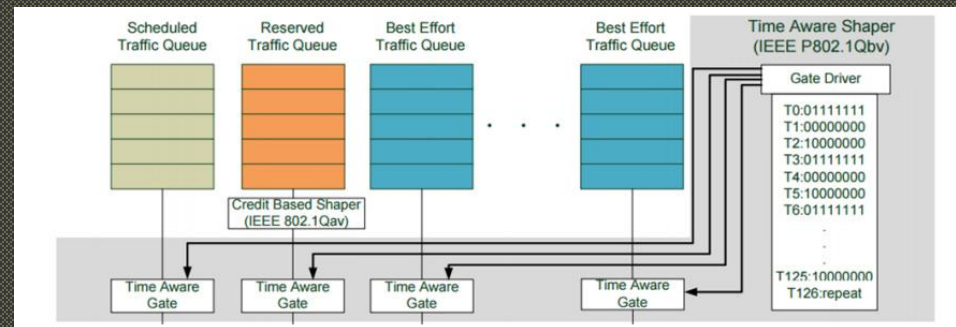
1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 CRITERIA OF LAN DESIGN

1. Support of Real-Time Control

Industrial automation relies on deterministic communication, where data packets must arrive within strict time limits (e.g., for robotic control, motion systems, and safety interlocks).

- Modern Solutions:
 - Time-Sensitive Networking (TSN): An extension of Ethernet that provides real-time, deterministic communication by prioritizing time-critical traffic.
 - Example: A PLC controlling a high-speed packaging line uses TSN to synchronize servo drives with microsecond precision.
- Industrial Ethernet Protocols:
 - PROFINET IRT (Isochronous Real-Time)
 - EtherNet/IP with CIP Sync
 - EtherCAT (for ultra-fast distributed control)



1.9 TYPES OF NETWORKS

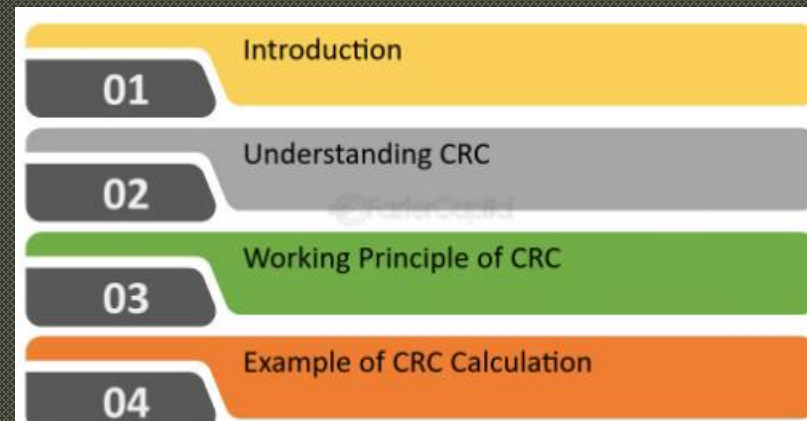
1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 CRITERIA OF LAN DESIGN

2. High Data Integrity (Error Detection & Correction)

Industrial networks must ensure zero data corruption, as even minor errors can cause production failures or safety hazards.

- Modern Solutions:
- Cyclic Redundancy Check (CRC) & Frame Validation: Enhanced error-checking mechanisms in Industrial Ethernet protocols.
- Packet Retransmission & Redundancy Protocols:
 - PRP (Parallel Redundancy Protocol) and HSR (High-availability Seamless Redundancy) for zero-failover in critical systems.
 - Example: A power substation automation system uses HSR to prevent communication failures during grid disturbances.



1.9 TYPES OF NETWORKS

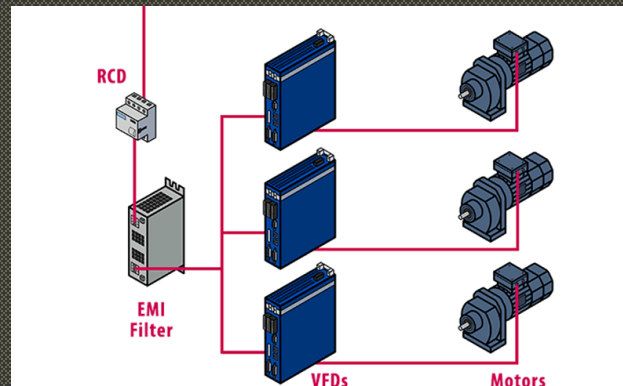
1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 CRITERIA OF LAN DESIGN

3. Freedom from High Noise (EMI/RFI Immunity)

Factories contain heavy machinery, variable-frequency drives (VFDs), and welding equipment that generate electromagnetic noise, which can disrupt network signals.

- Modern Solutions:
 - Shielded Twisted Pair (STP) & Fiber Optics:
 - CAT 6A/7 shielded cables for EMI resistance in motor control cabinets.
 - Fiber-optic backbone for noise-free long-distance communication.
 - M12 Connectors (IP67-rated): Ruggedized Ethernet connectors that prevent signal degradation in dusty/wet environments.
 - Example: An automotive assembly plant uses fiber-optic links between welding robots and the control room to avoid EMI-induced errors.



<https://www.kebamerica.com/blog/electromagnetic-interference-vfds/>

1.9 TYPES OF NETWORKS

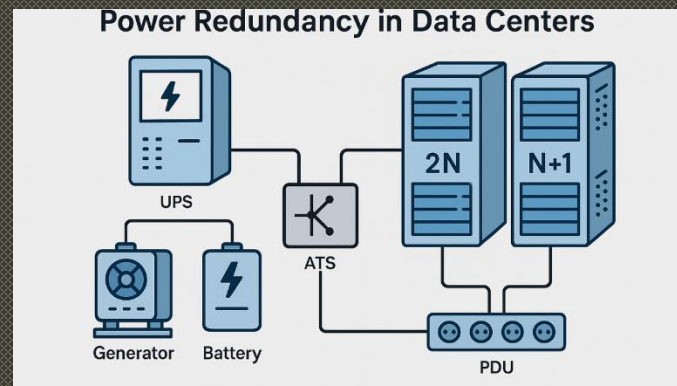
1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 CRITERIA OF LAN DESIGN

4. High Reliability (24/7 Uptime)

Industrial networks must operate continuously without downtime, as even brief interruptions can cost thousands in lost production.

- Modern Solutions:
 - Redundant Network Topologies:
 - Ring Topology (with Rapid Spanning Tree Protocol - RSTP) for fast failover (sub-50ms recovery).
 - Dual-star topologies for critical systems like oil & gas pipelines.
 - Industrial-Grade Switches:
 - Managed Ethernet switches with QoS (Quality of Service) to prioritize critical traffic.
 - Example: A pharmaceutical plant uses a self-healing fiber ring to ensure uninterrupted batch process control.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 CRITERIA OF LAN DESIGN

5. Scalability for Large Installations

Industrial LANs must support hundreds to thousands of devices across vast facilities (e.g., refineries, airports, smart factories).

- Modern Solutions:
 - Modular & Hierarchical Network Design:
 - Cell/Area Zone Architecture (per IEC 62443 standards) segregates networks for security and performance.
 - Distributed Control Nodes reduce latency in large plants.
 - Industrial Wireless (Wi-Fi 6, Private 5G):
 - For mobile equipment (AGVs, drones) in smart warehouses.
 - Example: A shipyard uses 5G-enabled LAN to coordinate autonomous cranes and logistics robots.



<https://www.csemag.com/ways-automation-and-scalability-are-shaping-industrial-facility-design/>

1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 FUNCTIONS OF LAN DESIGN

1. Communications Among Controllers and Other Stations

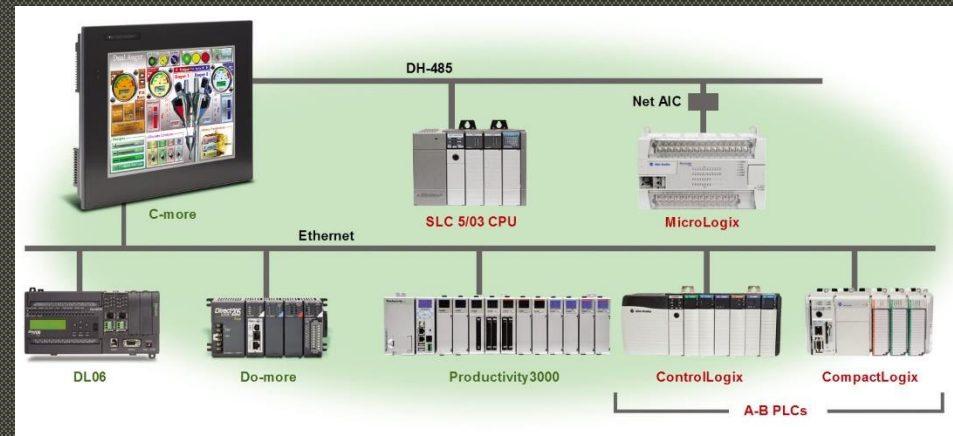
The LAN must enable seamless data exchange between PLCs, HMIs, SCADA systems, drives, and I/O modules.

Implementation & Technologies:

- Industrial Ethernet Protocols (PROFINET, EtherNet/IP, EtherCAT) for deterministic communication.
- OPC UA for standardized data exchange between different vendors' devices.
- Distributed Control Architecture where multiple PLCs share data (e.g., a bottling line where fillers, cappers, and labelers synchronize via LAN).

Example:

In an automotive assembly plant, robotic cells (PLCs) communicate with welding controllers and conveyor systems over a PROFINET LAN to ensure synchronized operations.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 FUNCTIONS OF LAN DESIGN

2. Uploading Programs from Any Controller to a Programming Device

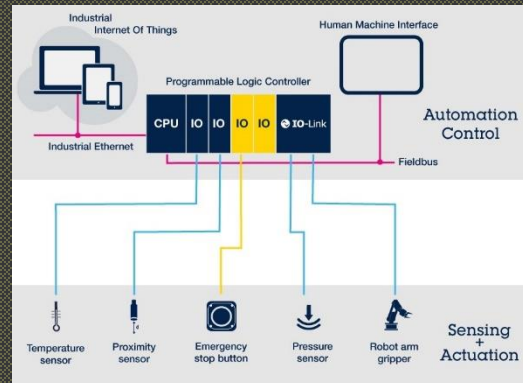
Engineers must be able to retrieve PLC logic, configurations, or firmware from any controller for backup, diagnostics, or version control.

Implementation & Technologies:

- Secure Remote Access via VPN or Tunneling Protocols (SSH, TLS).
- Centralized Engineering Stations (e.g., Siemens TIA Portal, Rockwell Studio 5000) with network-wide access.
- Role-Based Access Control (RBAC) to prevent unauthorized uploads.

Example:

A maintenance technician connects a laptop to a switch and uploads the latest PLC-5 program from a legacy motor control panel for troubleshooting.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 FUNCTIONS OF LAN DESIGN

3. Downloading Programs from a Programming Device to Any Controller

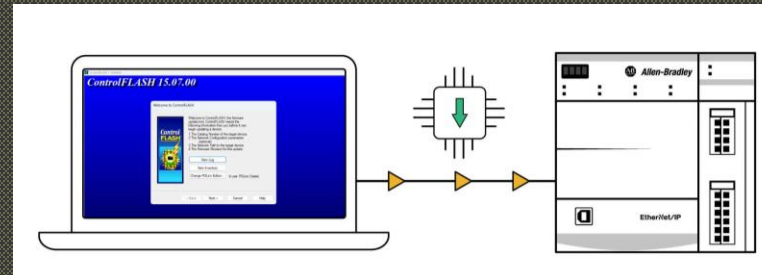
The LAN must support remote firmware updates, logic modifications, or configuration changes without physical access to each controller.

Implementation & Technologies:

- Secure Boot & Code Signing to prevent malicious updates.
- Delta Downloads (only sending changed logic blocks) to minimize downtime.
- Redundancy Protocols (PRP/HSR) to prevent corruption during updates.

Example:

An engineer pushes a new safety routine to multiple Allen-Bradley ControlLogix PLCs over EtherNet/IP without stopping production.



<https://www.realpars.com/blog/update-plc-firmware>

1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 FUNCTIONS OF LAN DESIGN

4. Reading/Writing I/O Values or Registers of Any Controller

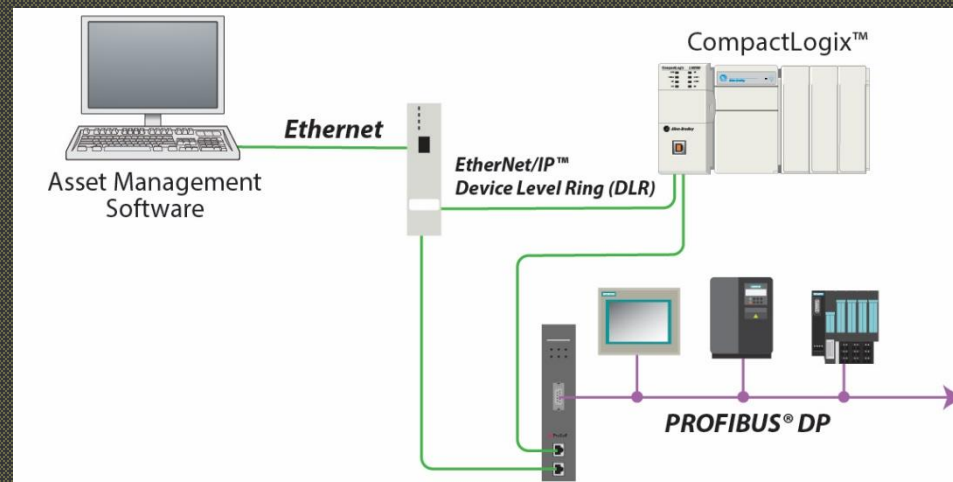
Real-time monitoring and control of I/O points, tags, and registers for process optimization and diagnostics.

Implementation & Technologies:

- Produced/Consumed Tags (in Rockwell systems) for shared memory between controllers.
- MQTT/Sparkplug B for IIoT-enabled data access.
- Web HMI/API Access (REST, OPC UA) for cloud-based monitoring.

Example:

A SCADA system reads temperature sensors (analog inputs) and adjusts PID loop setpoints in a Siemens S7-1500 PLC via PROFINET.



1.9 TYPES OF NETWORKS

1.9.1 LOCAL AREA NETWORK (LAN)

1.9.1.1 FUNCTIONS OF LAN DESIGN

5. Monitoring Controller Status & Controlling Operation

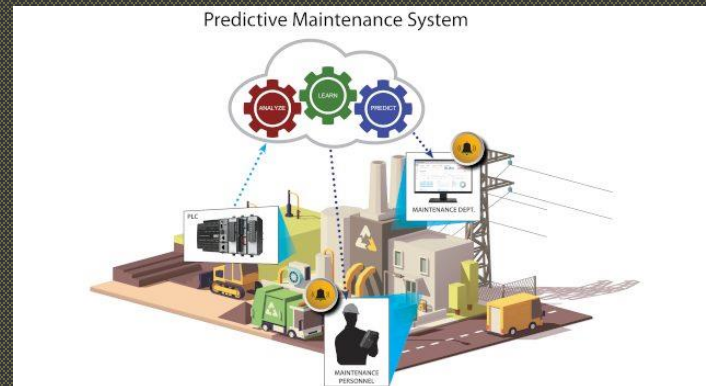
The LAN must provide real-time diagnostics, fault detection, and remote control (start/stop, mode changes).

Implementation & Technologies:

- SNMP & Network Management Software (e.g., Wireshark, PRTG) for traffic analysis.
- Predictive Maintenance Alerts via PLC heartbeat monitoring.
- Remote Stop/Start Commands (with safety authentication).

Example:

A plant operator detects a stuck valve via HMI, forces an output to reset it, and checks the controller's diagnostic buffer for errors - all over the LAN.

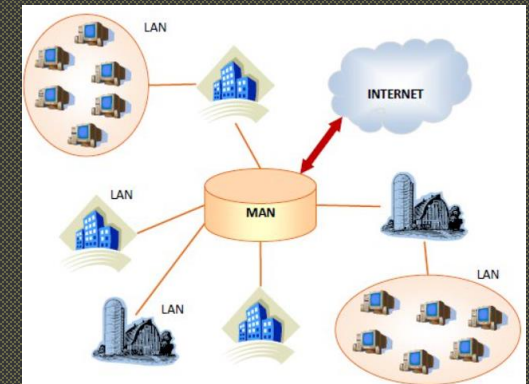


<https://www.controleng.com/plcs-improve-predictive-maintenance/>

1.9 Types of Networks

1.9.2 Metropolitan Area Network (MAN)

- A Metropolitan Area Network (MAN) is a high-capacity network that spans a city or large campus (typically under 50 km), filling the gap between Local Area Networks (LANs) and Wide Area Networks (WANs). MANs are designed to interconnect multiple LANs across a metropolitan region, providing high-speed data, voice, and video services to businesses, universities, and government agencies.
- Geographical Coverage:
 - Smaller than WANs (which can span countries/continents) but larger than LANs (limited to a building or campus).
 - Typical range: 5 km to 50 km (e.g., connecting corporate offices across a city).
- Speed and Performance:
 - 56 kbps to 100 Mbps (traditional MANs).
 - Modern fiber-based MANs now support 1 Gbps to 100 Gbps (e.g., municipal broadband networks).
 - Faster than WANs (due to shorter distances) but slower than high-speed LANs (10 Gbps+).
- Technology Used:
 - Fiber-optic backbone (single-mode or multi-mode) for high bandwidth and low latency.
 - Microwave/Wireless MAN (WiMAX, 5G mmWave) for flexible deployments.
 - MPLS (Multiprotocol Label Switching) for efficient traffic routing.



<https://www.quora.com/What-is-metropolitan-area-network-MAN>

1.9 Types of Networks

1.9.2 Metropolitan Area Network (MAN)

1. Smart City Infrastructure

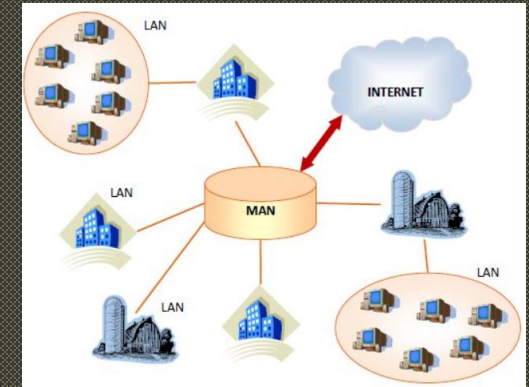
- A city deploys a fiber-optic MAN to connect:
- Traffic light controllers (for adaptive signal timing).
- Surveillance cameras (real-time video analytics).
- Public Wi-Fi hotspots (free internet access in downtown areas).
- Technology Used: Dark fiber leases + Carrier Ethernet.

2. Telecom Service Providers (ISP Backbone)

- An ISP uses a MAN to:
- Aggregate customer connections (FTTH, business leased lines). **FTTH – Fiber To The Home**
- Peer with other ISPs at metropolitan exchange points.
- Technology Used: DWDM (Dense Wavelength Division Multiplexing) for maximizing fiber capacity.

3. Industrial Automation (Multi-Plant Connectivity)

- A manufacturing company links:
- Factory LANs (PLCs, SCADA systems).
- Warehouse networks (inventory tracking).
- Corporate HQ (ERP integration). **ERP – Enterprise Resource Planning**
- Technology Used: Private 5G MAN for low-latency machine communications.

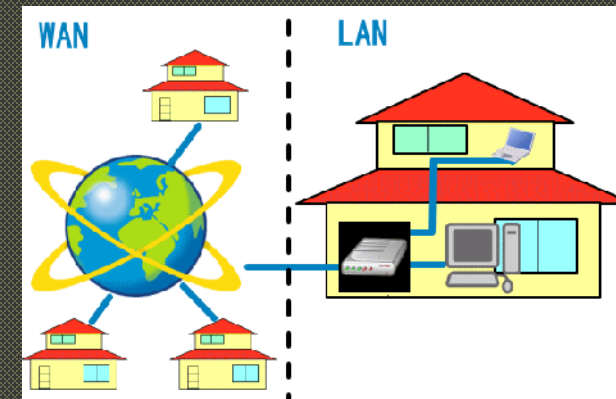


<https://www.quora.com/What-is-metropolitan-area-network-MAN>

1.9 Types of Networks

1.9.3 Wide Area Network (WAN)

- A Wide Area Network (WAN) is a telecommunications network that spans large geographical distances - from a few kilometers to thousands of kilometers - connecting multiple Local Area Networks (LANs) or other networks. Unlike LANs, which are confined to a single building or campus, WANs enable global communication between offices, data centers, and remote sites.
- The most well-known example of a WAN is the Internet, but private WANs are also widely used by enterprises, governments, and service providers.
- Range: From a few kilometers (multi-site campuses) to global coverage (multinational corporations).
- Examples:
 - A bank connecting branches across different cities.
 - A global retail chain linking stores, warehouses, and HQ.



1.9 Types of Networks

1.9.3 Wide Area Network (WAN)

Reliability & Error Handling

- Challenge: WANs are less reliable than LANs due to distance, congestion, and varying connection quality.
- Solutions:
 - Error Detection/Correction: TCP retransmission, FEC (Forward Error Correction).
 - Redundancy: Dual ISP links, failover routing.
 - QoS (Quality of Service): Prioritizing voice/video traffic.

Communication Methods

- Circuit Switching (Legacy): Dedicated path (e.g., PSTN, ISDN).
- Packet Switching (Modern): Data split into packets (e.g., MPLS, Internet).
- SD-WAN: Dynamically routes traffic over multiple links (e.g., broadband + LTE backup).

Technology	Speed	Use Case
Fiber Optics	1 Gbps – 100 Gbps	High-speed enterprise backbones
MPLS	10 Mbps – 10 Gbps	Reliable corporate networks
4G/5G Wireless	50 Mbps – 1 Gbps	Remote site & IoT connectivity
Satellite Links	1 Mbps – 100 Mbps	Oil rigs, ships, rural areas
Leased Lines	64 Kbps – 10 Gbps	Dedicated secure connections

- PSTN - Public Switched Telephone Network, is the traditional telephone network.
- ISDN - Integrated Services Digital Network, is a digital network designed to carry voice, data, and video over the same line.
- LTE - Long Term Evolution, wireless communication standard.

SD-WAN - Software-Defined WAN
MPLS - Multiprotocol Label Switching

1.9 Types of Networks

1.9.3 Wide Area Network (WAN)

- Enterprise Networks (Private WANs)
 - A multinational corporation uses an MPLS WAN to connect:
 - HQ (New York) → Manufacturing plant (Germany) → Call center (India).
 - Tech Used: SD-WAN + VPN for secure, low-cost connectivity.

2. Cloud & Data Center Connectivity

- AWS Direct Connect provides private fiber links between a company's data center and cloud providers.
- Speed: 1 Gbps – 100 Gbps dedicated connections.

3. Retail & Banking Networks

- A bank uses a WAN to:
- Sync transactions between branches in real-time.
- Process ATM withdrawals across different regions.
- Tech Used: Encrypted leased lines + failover 4G/5G.

4. Industrial & IoT WANs

- Smart grid systems use WANs to:
- Monitor power distribution across cities.
- Control remote wind farms via cellular (LTE-M/NB-IoT).

Feature	LAN	MAN	WAN
Range	< 1 km (building/campus)	< 50 km (citywide)	Global (unlimited)
Speed	1 Gbps – 100 Gbps	100 Mbps – 100 Gbps	1 Mbps – 10 Gbps
Ownership	Enterprise/Private	ISP/Government	Telecom Carriers
Topology	Star/Ring (Ethernet)	Ring/Mesh (MPLS, SONET)	Mesh (Internet, SD-WAN)
Latency	Ultra-low (<1 ms)	Low (2-10 ms)	High (50-300 ms)

Failover 4G/5G" refers to using a 4G or 5G cellular connection as a backup or secondary internet connection

- LTE-M - Long Term Evolution for Machines
- NB-IoT (Narrowband IoT)
- Both designed for low-power, wide-area (LPWA) Internet IoT

1.9 Types of Networks

1.9.4 Personal Area Network (PAN)

- A Personal Area Network (PAN) is a small-scale wireless network designed for short-range communication between personal devices, typically within 30 feet (10 meters).
- Unlike LANs (Local Area Networks) or WANs (Wide Area Networks), PANs are optimized for low-power, low-cost, and seamless connectivity between gadgets like smartphones, wearables, and peripherals.
- The most common PAN technology is Bluetooth, but alternatives like Infrared (IrDA) and Near Field Communication (NFC) are also used in specific cases.

- Zigbee and Bluetooth are both wireless technologies.
- Zigbee is designed for low-power, low-data-rate applications like smart home automation and industrial sensor networks.
- Bluetooth is commonly used for short-range, higher-data-rate connections between devices like headphones and speakers.

Technology	Range	Speed	Common Use Cases
Bluetooth	Up to 30 ft	1-3 Mbps (Classic) 50 Mbps (Bluetooth 5+)	Wireless earbuds, keyboards, smartwatches
Infrared (IrDA)	~3 ft (Line-of-sight)	4 Mbps max	Older TV remotes, PDA syncing
NFC	< 4 inches	424 Kbps	Contactless payments, device pairing
Zigbee	30-300 ft (low-power)	250 Kbps	Smart home sensors, IoT devices

PDA - Personal Digital Assistant

1.9 Types of Networks

1.9.5 Wireless Local Area Network (WLAN)

- A Wireless Local Area Network (WLAN) is a network that connects devices wirelessly within a limited area (such as a home, office, or campus) using Wi-Fi technology.
- Unlike traditional wired LANs that rely on Ethernet cables, WLANs provide flexible, high-speed connectivity without physical constraints.
- The term "Wi-Fi" (Wireless Fidelity) refers to wireless networking technologies based on the IEEE 802.11 standards. These standards define how wireless devices communicate, ensuring compatibility across manufacturers.

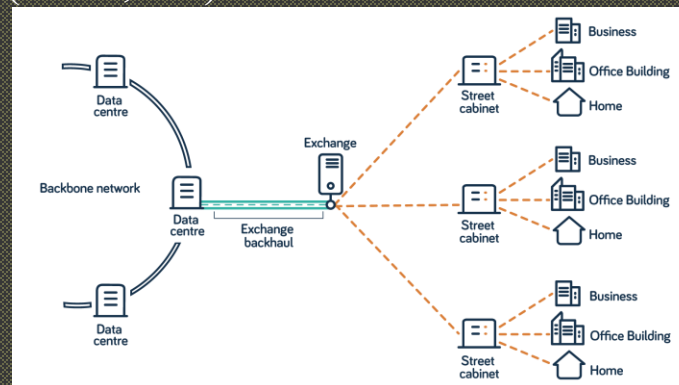
Standard	Release Year	Max Speed	Frequency Band	Key Features
802.11a	1999	54 Mbps	5 GHz	First 5 GHz Wi-Fi, less interference
802.11b	1999	11 Mbps	2.4 GHz	Popularized Wi-Fi in homes
802.11g	2003	54 Mbps	2.4 GHz	Faster than 802.11b, backward compatible
802.11n (Wi-Fi 4)	2009	600 Mbps	2.4 GHz & 5 GHz	MIMO (Multiple-Input Multiple-Output)
802.11ac (Wi-Fi 5)	2013	3.5 Gbps	5 GHz	Wider channels (160 MHz), Beamforming
802.11ax (Wi-Fi 6)	2019	9.6 Gbps	2.4 GHz & 5 GHz	OFDMA, BSS Coloring, better efficiency
802.11be (Wi-Fi 7)	2024	40 Gbps	2.4/5/6 GHz	320 MHz channels, Multi-Link Operation

1.9 Types of Networks

1.9.5 Wireless Local Area Network (WLAN)

Components:

1. Wireless Access Points (WAPs)
 - Function: Broadcast Wi-Fi signals to devices.
 - Example: A corporate office uses multiple enterprise-grade APs (like Cisco Aironet) for full coverage.
2. Wi-Fi Routers
 - It combines a WAP (for Wi-Fi), a firewall (for security), and a switch (to connect multiple devices) into one device, ideal for homes or small offices.
 - Example: Google Nest Wifi provides mesh networking for seamless roaming.
3. Client Devices
 - Laptops, smartphones, IoT gadgets with Wi-Fi adapters.
4. Backhaul Connection
 - Wired (Ethernet/fiber) or wireless (mesh, 5G) link to the internet.



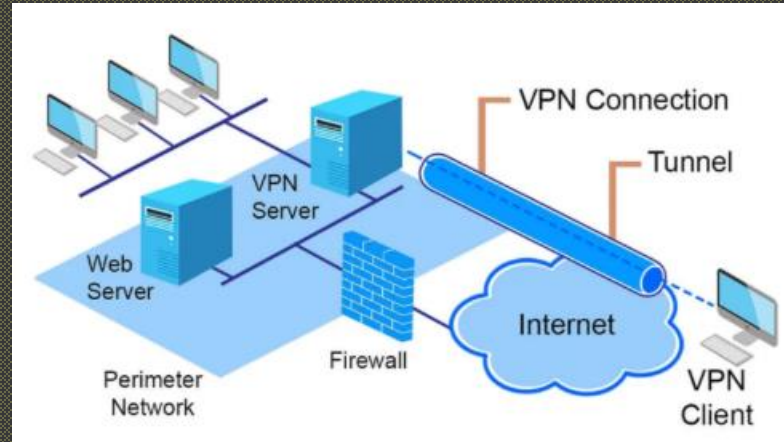
A backhaul connection is a high-capacity link connecting local networks or subnetworks to a central or backbone network.

1.9 Types of Networks

1.9.7 Virtual Private Networks (VPN)

A Virtual Private Network (VPN) lets organizations use the public Internet to create a secure, private network. It creates a secure, private connection that hides your IP address and encrypts your data, keeping your online activity safe from tracking by your ISP, websites, or others. It is especially useful on public Wi-Fi for private and secure browsing.

- Private: Data is encrypted, so outsiders can't read it.
- Virtual: It uses the Internet (a public network) instead of expensive private cables.



1.9 Types of Networks

1.9.7 Virtual Private Networks (VPN)

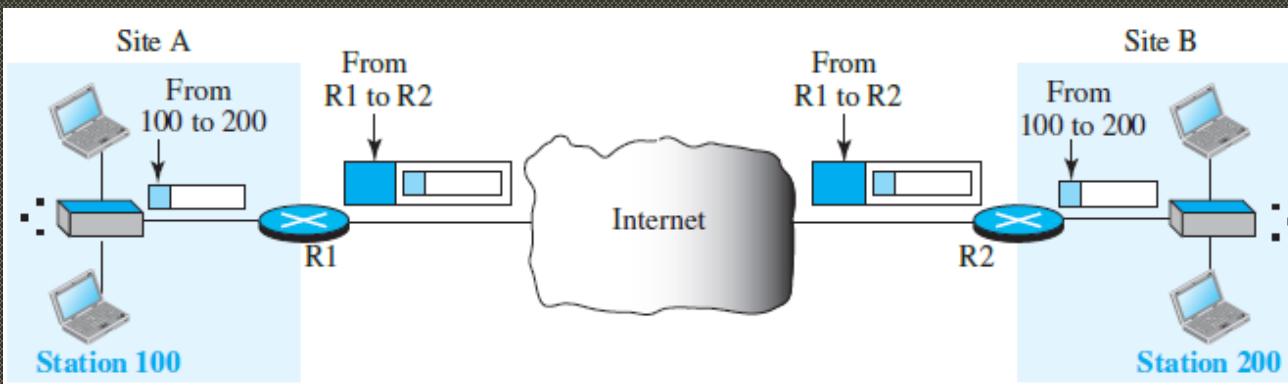
1. Two Sites (Site A & Site B) need to communicate securely over the Internet.
2. Routers R1 & R2 act as VPN gateways:
 - R1 (Site A) encrypts data before sending it.
 - R2 (Site B) decrypts and forwards it to the right device.
3. IPsec in Tunnel Mode (for maximum security):
 - The original private packet (from Station 100 to Station 200) is fully encrypted.
 - A new outer packet is created with R1's & R2's IP addresses (so the Internet can route it).
 - Outsiders see only R1 → R2 traffic, not the real sender/receiver (100 → 200).
4. At Site B (R2):
 - The outer packet is removed.
 - The inner packet is decrypted and sent to Station 200.

IPsec is a security protocol suite that encrypts and authenticates data sent over IP networks. It ensures:

- Confidentiality (no one can read your data).
- Integrity (no one can alter your data).
- Authentication (verifies who sent the data).

IPsec operates in two main modes:

- Tunnel Mode
 - Encrypts the entire original packet (header + data).
 - Adds a new IP header (like a "fake address") for routing.
 - Used in VPNs to hide internal networks.
- Transport Mode
 - Encrypts only the data (keeps original IP headers).
 - Used for direct device-to-device security (e.g., server-to-server).

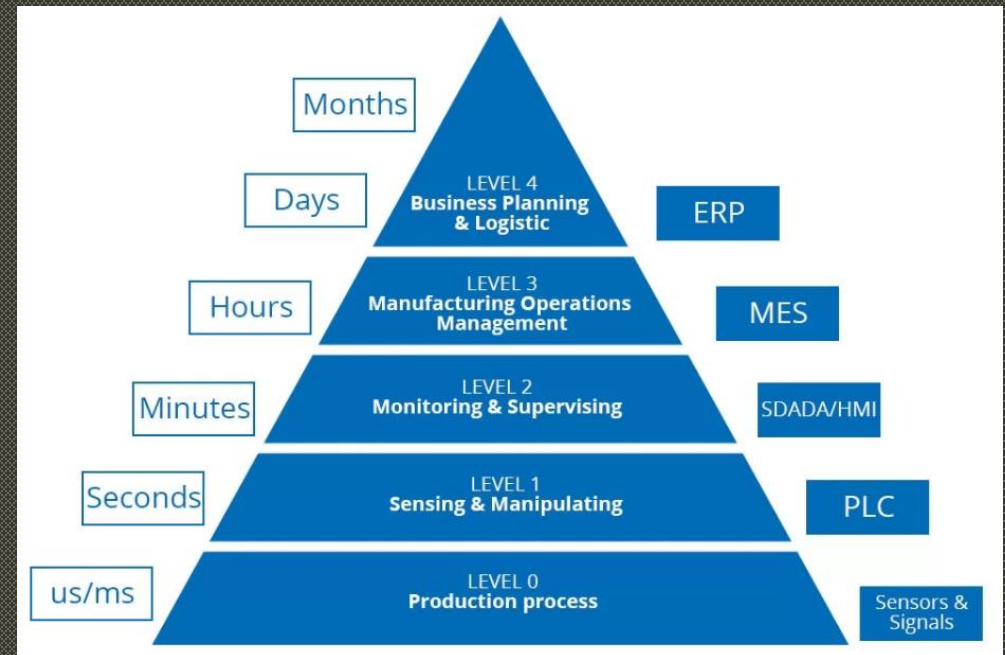


1.10 Hierarchy of Industrial Automation Systems

The industrial automation hierarchy is a structured framework that organizes control and management functions into distinct levels. Each layer plays a critical role in ensuring efficient, reliable, and intelligent manufacturing operations.

1. Level 0: Field Devices (Sensors & Actuators) - Direct interaction with physical processes

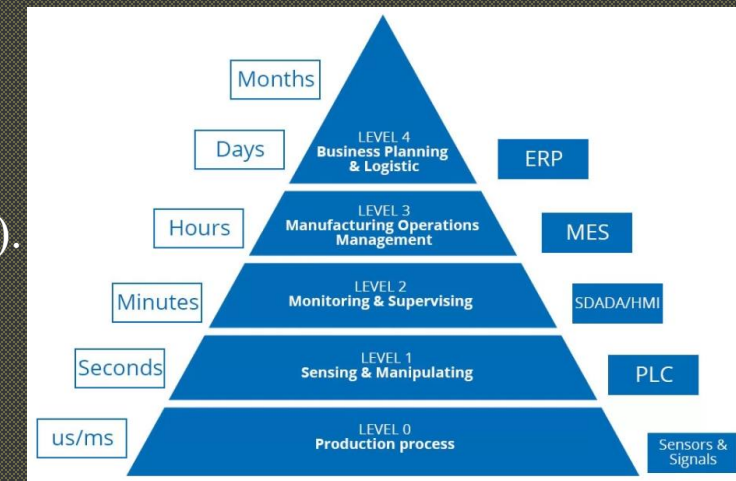
- Components:
 - Sensors (temperature, pressure, flow meters)
 - Actuators (motors, valves, relays)
 - I/O Modules (Analog/Digital signals)
- Function:
 - Collect real-time data (e.g., "Motor temperature = 75°C").
 - Execute commands (e.g., "Open valve to 50%").
- Example:
 - A pressure sensor in an oil refinery detects a leak and triggers an emergency shutdown.



1.10 Hierarchy of Industrial Automation Systems

2. Level 1: Basic Control (PLCs & PID Controllers) - Real-time machine/process regulation

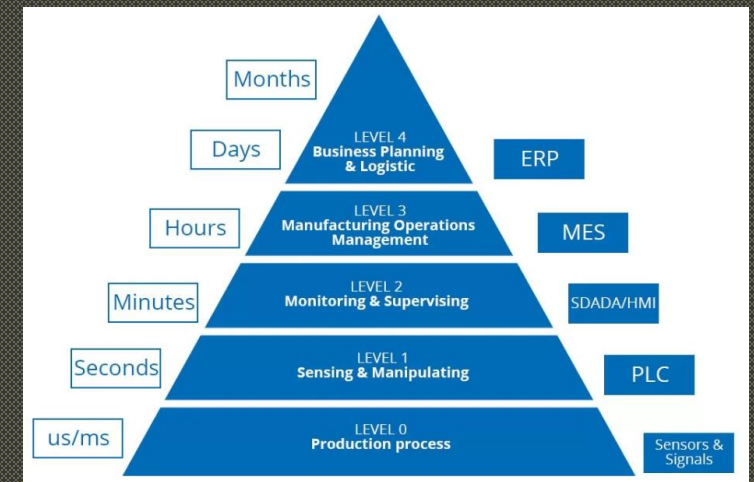
- Components:
 - PLCs (Programmable Logic Controllers)
 - PID Controllers (for precision tuning)
 - Edge Devices (local data processing)
- Function:
 - Run logic-based automation (e.g., "If temperature $> 100^{\circ}\text{C}$, turn on cooling fan").
 - Maintain setpoints (e.g., "Keep pH level at 7.0").
- Example:
 - A PLC in a bottling plant controls conveyor speed based on sensor inputs.



<https://excelpro.ca/en/news/the-automation-pyramid-isa-95>

1.10 Hierarchy of Industrial Automation Systems

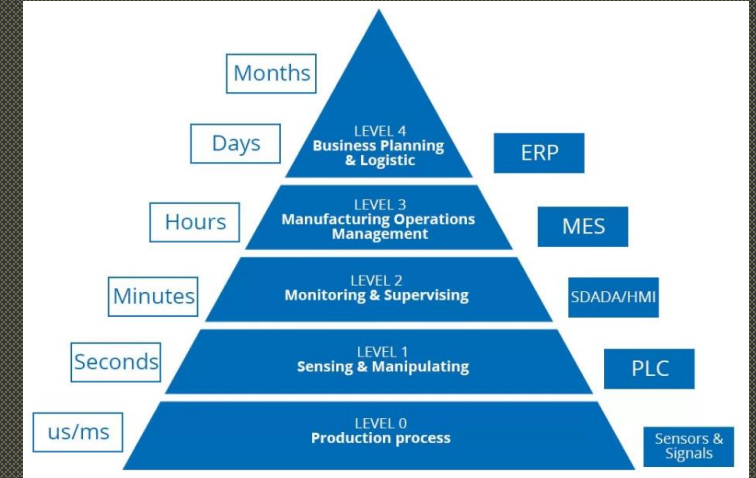
3. Level 2: Supervisory Control (SCADA & HMI) - Monitoring & coordination of multiple processes
- Components:
 - SCADA Systems (Supervisory Control and Data Acquisition)
 - DCS (Distributed Control Systems)
 - HMIs (Human-Machine Interfaces)
 - Function:
 - Visualize plant operations (dashboards, alarms).
 - Optimize batch processes (e.g., "Adjust reactor pressure across 5 units").
 - Example:
 - A SCADA system in a water treatment plant tracks reservoir levels and triggers pumps as needed.



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1.10 Hierarchy of Industrial Automation Systems

- 4. Level 3: Manufacturing Execution System (MES) - Bridging operations & business planning
 - Components:
 - MES Software (Siemens Opcenter, Rockwell MES)
 - Real-Time Production Tracking
 - Quality Management Systems
 - Function:
 - Schedule production orders.
 - Track OEE (Overall Equipment Effectiveness).
 - Enforce compliance (e.g., FDA regulations in pharma).
 - Example:
 - A MES in an automotive plant schedules robot welding tasks to meet daily quotas.

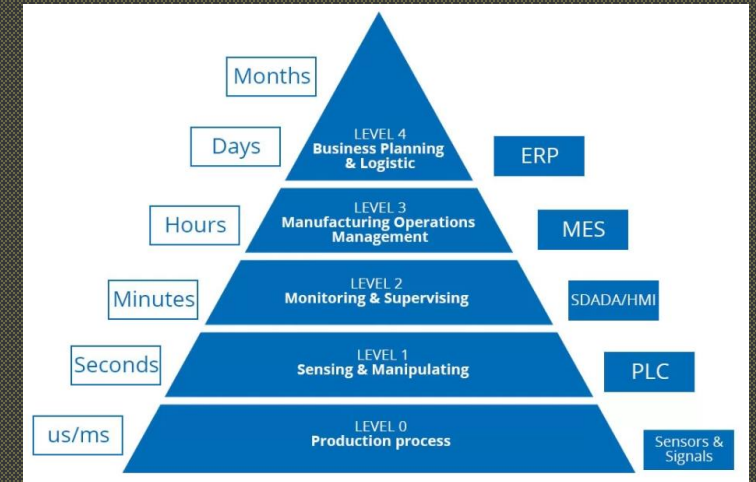


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1.10 Hierarchy of Industrial Automation Systems

5. Level 4: Enterprise Resource Planning (ERP) - Business-wide decision-making

- Components:
 - ERP Systems (SAP, Oracle)
 - AI-Driven Analytics
 - Cloud Integration
- Function:
 - Manage supply chain, inventory, finances.
 - Predict demand using machine learning.
- Example:
 - An ERP analyzes global sales data to adjust raw material orders.



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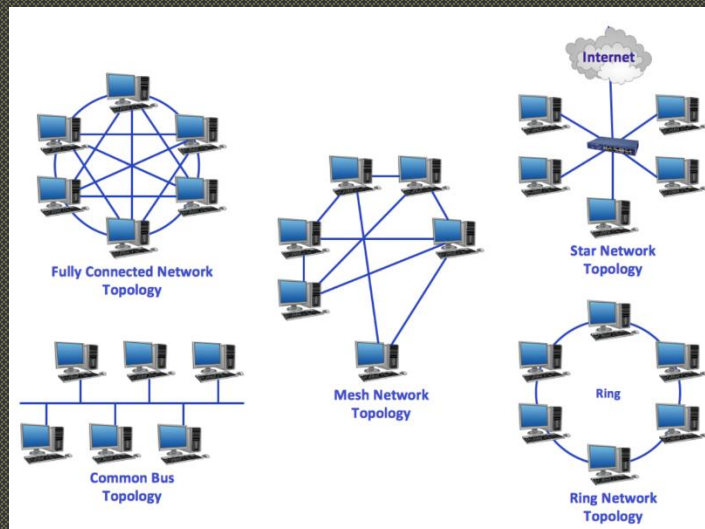
Beyond the Pyramid: Industry 4.0 & IIoT

Modern automation integrates new technologies that blur traditional hierarchy lines:

- IIoT (Industrial Internet of Things):
 - Smart sensors send data directly to cloud AI.
- Digital Twins:
 - Virtual replicas of physical systems for simulation.
- AI/ML:
 - Predictive maintenance (e.g., "Bearing failure likely in 48 hrs").

1.11 Network Topologies

- Network Topologies describe both the physical layout of cabling/pathways and the logical flow of data between devices.
- Network Technologies/Architectures define how computers transmit data to the network medium in an organized way.
- Certain technologies are compatible only with specific topologies, limiting viable combinations.
 - Attempting to use Token Ring with a star topology (like modern Ethernet) would be incompatible. The ring topology was essential because Token Ring's protocol relied on the token-passing mechanism, which wouldn't function in a star or bus topology.
- Understanding older, now-obsolete technologies provides context for current and future network advancements.
- Wi-Fi (802.11 Standards) is rapidly evolving, often replacing wired Ethernet in some environments as the preferred connection method.



• **Physical Topology:** Arrangement of cabling and device connections in a network.

- Example: A small office with Ethernet cables connecting computers to a central switch in a star pattern.

• **Logical Topology:** The path data takes between computers, regardless of physical layout.

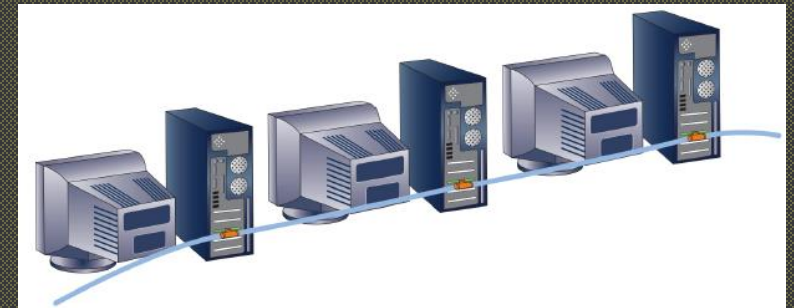
- Example: In the same office, data travels as if all computers are on a single cable (bus logical topology).

1.11 Network Topologies

1.11.1 Bus Topology

The physical bus topology is the simplest and most common way to connect computers in a network.

- A single continuous cable connects computers in a daisy-chain fashion.
- New computers could be added by extending the cable from the last node.
- Advantages
 - Simple setup – Easy to install and expand.
 - Low cost – Requires minimal cabling.
- Disadvantages
 - Limited scalability – Only 30 computers per segment allowed.
 - Short distance – Max 185 meters of total cable length.
 - Fragile connectivity – Any break in the cable shuts down the whole network.
 - Disruptive maintenance – Adding/removing a computer temporarily stops the network.
 - Slow speeds – Limited to 10 Mbps half-duplex (due to coaxial cabling).
 - Termination required – Both ends of the bus must be properly terminated to prevent signal reflection.

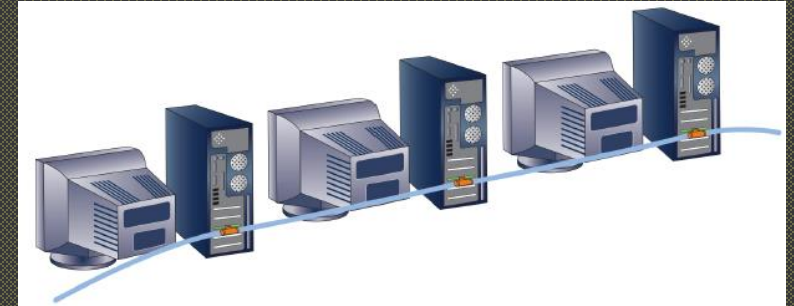


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1.11 Network Topologies

1.11.1 Bus Topology

- Signal Propagation
 - Data travels as electrical pulses along a copper bus cable in all directions.
 - Signals weaken over distance or get absorbed by devices.
- Signal Bounce (Reflection)
 - If a signal hits the end of an unterminated cable, it reflects back, causing interference (like an echo).
 - This disrupts subsequent signals, corrupting data.
- Solution: Termination
 - Terminators (resistors) are attached to both ends of the bus cable.
 - They absorb signals instead of letting them bounce back.
 - Prevents signal interference, ensuring clean communication.
- Real-World Analogy
 - Imagine shouting in an empty, echoey room - your words get distorted.
 - Adding sound-absorbing foam (like a terminator) stops echoes, making speech clear.
- Early Ethernet (10BASE5/10BASE2) relied on proper termination.
- Modern networks (switched Ethernet) avoid this issue with point-to-point links.



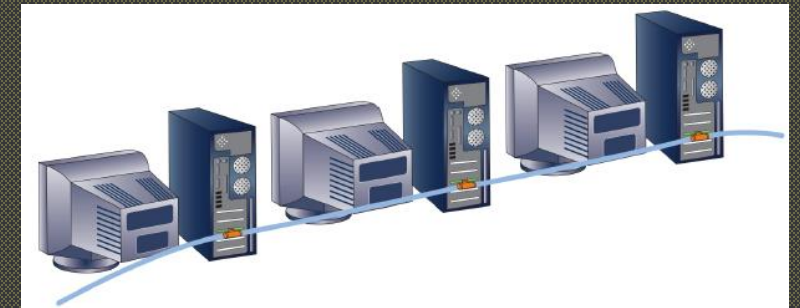
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1.11 Network Topologies

1.11.1 Bus Topology

Limitations

- 30-Device Limit per Segment: Each connected workstation weakens the signal until it becomes unreadable after 30 nodes.
- 185-Meter Cable Limit: Signal degradation restricts total cable length, regardless of how many devices are attached.
 - Solution: Use a repeater to regenerate the signal (extends range but adds complexity).
- Termination Required
 - Both ends must have terminators to prevent signal bounce (reflection causing data corruption).
 - Adding/removing a device disrupts termination, temporarily breaking the network.
- Slow Speed (10 Mbps Half-Duplex)
 - Coaxial cables (like cable TV wires) limit bandwidth and force one-way communication at a time.
 - Modern twisted-pair cables (e.g., Cat6) support 1,000+ Mbps full-duplex.
- Bus topology is not the preferred choice nowadays.
- Star topology (centralized switches) replaced buses.
- Early Ethernet (10BASE2/10BASE5) used bus topology.



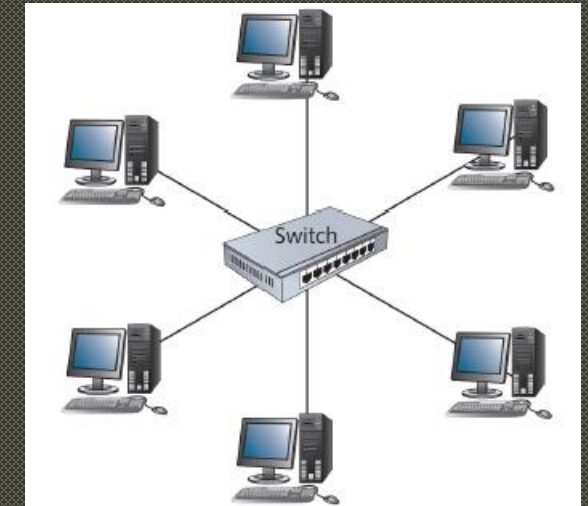
1.11 Network Topologies

1.11.2 Star Topology

- Uses a central device (hub/switch) to connect all computers in a LAN via individual cables.

Advantages

- Faster Speeds
 - Supports Gigabit (1000 Mbps+) full-duplex communication (vs. 10 Mbps half-duplex in bus topology).
 - Central device (e.g., a switch) enables simultaneous two-way data transfer.
- Centralized Management
 - Monitors traffic, detects errors, and alerts admins (in enterprise-grade switches).
 - Enables traffic statistics and performance analysis.
- Easy Upgrades
 - Upgrade only the central device (e.g., replace a hub with a switch) for faster speeds.
 - NICs can be upgraded gradually (mixed-speed support).
- Scalability: Expand by connecting multiple switches (forming an extended star).
- Dominates modern LANs
- No single point of failure (unlike bus topology's cable breaks).
- Hot-swappable - adding/removing devices doesn't disrupt the network.
- Flexible - supports mixed speeds (10/100/1000 Mbps) on the same switch.



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1.11 Network Topologies

1.11.2 Star Topology

An extended star topology (or hierarchical star) connects multiple star networks through a central high-speed switch, creating a scalable and efficient LAN structure.

- Central Core: A high-speed switch (e.g., 1000 Mbps) acts as the main hub.
- Outer Layers: Additional switches (not computers) connect to the core, forming secondary stars.
- End Devices: Computers/peripherals attach to the outer switches.

Advantages

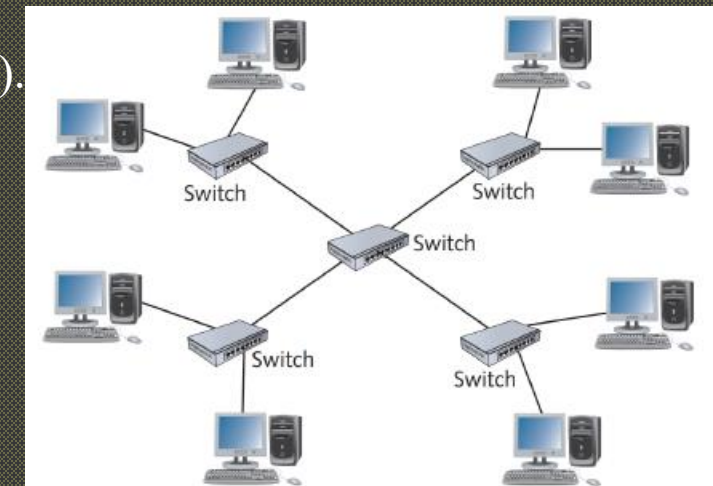
- Scalability:
 - Supports hundreds of devices by adding more switches.
- Performance Optimization
 - Core switch runs at higher speeds (e.g., 1000 Mbps) than edge devices (e.g., 100 Mbps).
- Traffic Management
 - Isolates local traffic (e.g., a department's traffic stays within its switch).
- Fault Tolerance
 - Failure in one branch doesn't collapse the entire network.

Standard for Large Networks

- Enterprise Offices: Connects floors/departments efficiently.
- Data Centers: Links server racks to a high-speed core.
- Campuses: Manages buildings under a unified network.

Example

- Humber uses an extended star:
- Core: 10 Gbps switch in the IT center.
- Layer 2: 1 Gbps switches in each building.
- Edge: 100 Mbps classrooms/labs.

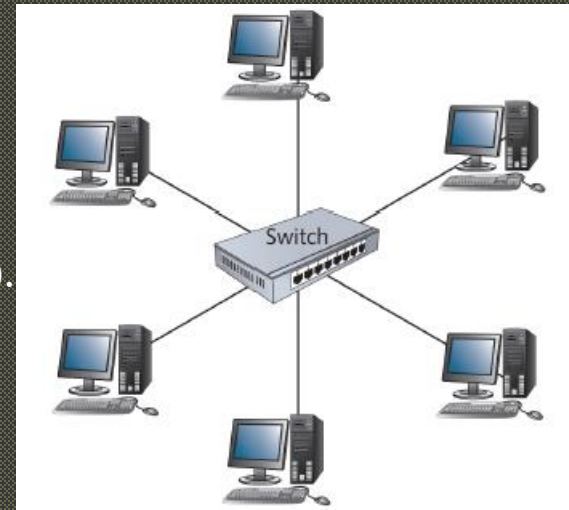


1.11 Network Topologies

1.11.2 Star Topology

- Data Transmission Process
 - Central Device Determines Path:
 - Hub: Broadcasts data to all connected devices (inefficient, shared bandwidth).
 - Switch: Directs data only to the intended recipient (efficient, dedicated bandwidth).
 - MAU (Token Ring): Uses a token-passing method (obsolete in modern networks).
- Disadvantages of Physical Star
- Single Point of Failure
 - If the central switch/hub fails, the entire network goes down.
 - Mitigation: Use redundant switches or UPS (uninterruptible power supply).
- Higher Cable Usage
 - Requires individual cables from each device to the central hub/switch (vs. bus topology's daisy-chain).
 - Initially it was a cost concern, but cable prices dropped, making star topology affordable.

MAU – Media Access Unit (Transceiver – Cable and NIC)



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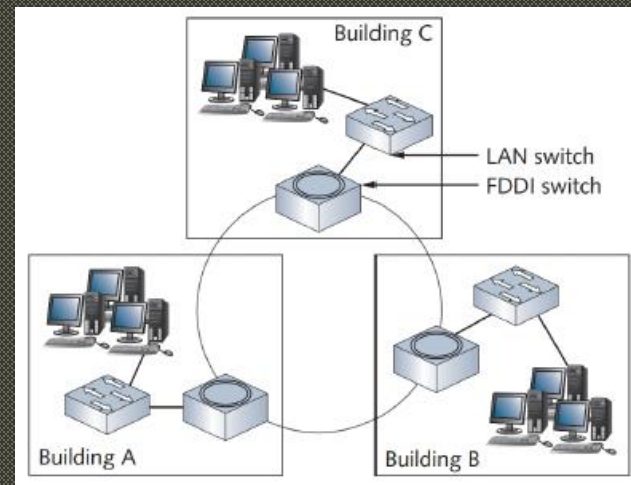
Star Topology Preferred Topology

- Speed: Switches enable Gigabit+ full-duplex communication.
- Reliability: Isolated link failures (e.g., one cable breaks) don't crash the whole network.
- Scalability: Easy to add devices without disruption.

1.11 Network Topologies

1.11.3 Ring Topology

- Devices are daisy-chained in a closed loop, with no termination points (unlike a bus).
- Data travels unidirectionally from one station to the next until it reaches its destination.
- Main Use
 - FDDI (Fiber Distributed Data Interface):
 - Provided high-speed backbone connections between LANs (e.g., linking buildings).
 - Used dual-ring redundancy for fault tolerance (data could reverse direction if a node failed).



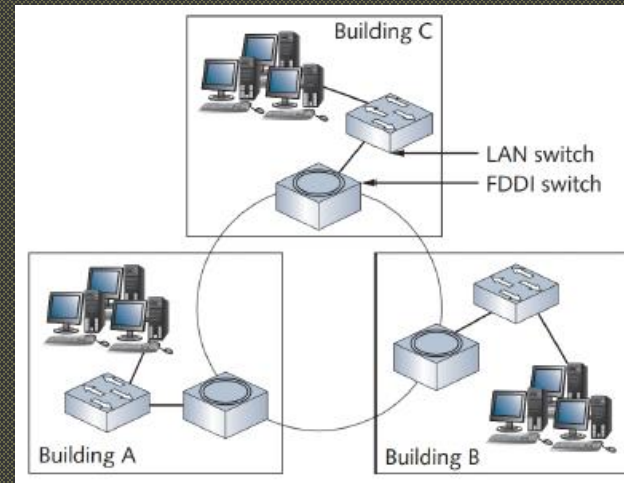
- FDDI was costly but provided 100 Mbps fiber-optic backbones in the 1990s.
- Modern networks use switched fiber (e.g., 10G Ethernet) instead.

1.11 Network Topologies

1.11.3 Ring Topology

Disadvantages

- Single Point of Failure:
 - If one device fails, the entire ring breaks (unless using FDDI's dual-ring backup).
- Slow & Inefficient:
 - Each station must reprocess and forward data, creating latency.
- Obsolete for LANs:
 - Replaced by faster, more reliable star/extended star topologies.
- Not the preferred topology - Ethernet's star topology offered:
 - Better scalability.
 - No dependency on every node being operational.
 - Higher speeds (e.g., Gigabit Ethernet).



1.11 Network Topologies

1.11.4 Point-to-Point Topology

A direct, dedicated connection between two devices, commonly used in:

- WANs/MANs (e.g., business-to-ISP links).
- Wireless bridges (connecting two buildings without cables).
- Ad-hoc file transfers (two computers linked directly via NICs – when needed exchange files).

Features

- Exclusive Bandwidth:
 - Data travels on a private link, avoiding shared network congestion.
- High Reliability:
 - Minimal interference or latency (ideal for critical connections).



1.11 Network Topologies

1.11.4 Point-to-Point Topology

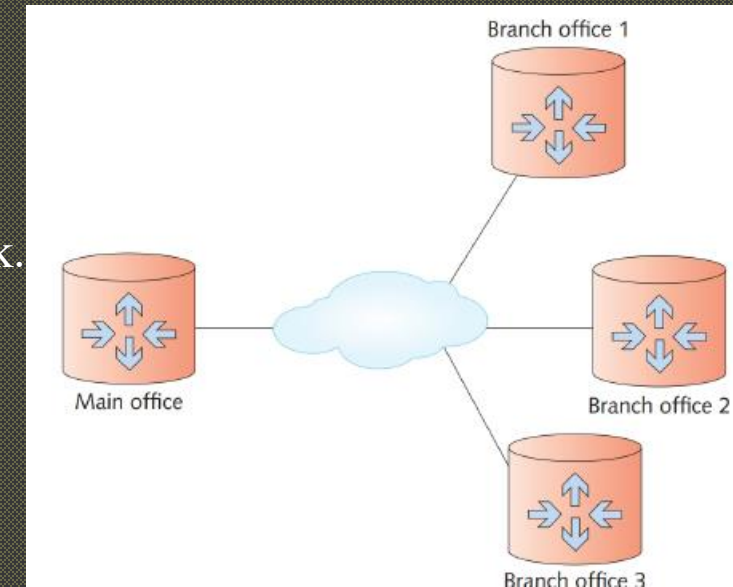
Point-to-Multipoint topology is a centralized communication model where a single hub device (e.g., router, base station) connects and manages multiple remote devices.

- Central Hub: Acts as the gateway for all communication.
- Remote Nodes: Branch offices, subscriber stations, or IoT devices.
- Traffic Flow: All data passes through the hub (no direct device-to-device communication).

Features

- Efficient for Hub-Spoke Networks:
 - Reduces cabling/WAN costs vs. multiple point-to-point links.
- Scalable:
 - Easy to add new branches/subscribers without reconfiguring the entire network.
- Common in Wireless:
 - Cellular towers (base stations) serve many devices.
 - Wi-Fi access points connect multiple clients.

- The "cloud" symbolizes the service provider's switching infrastructure.
- Example: A corporate router → ISP cloud → branches in different cities.



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1.11 Network Topologies

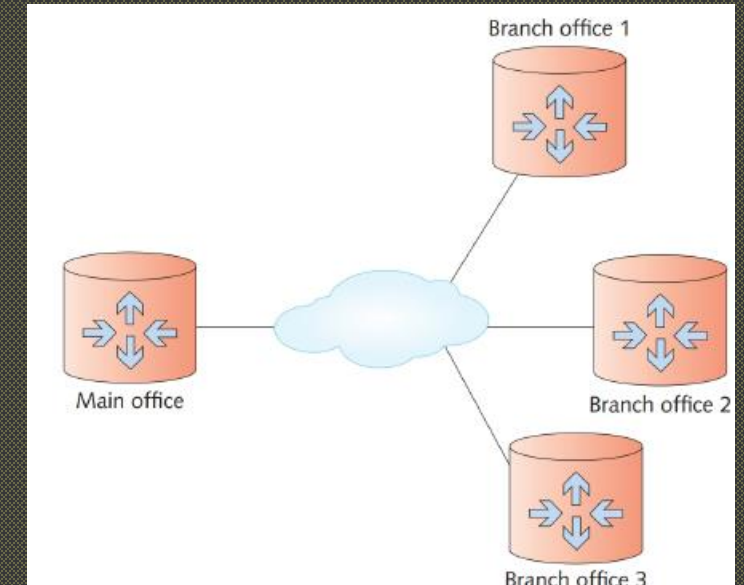
1.11.4 Point-to-Point Topology

Applications:

- Enterprise WANs:
 - HQ router links to branch offices via a service provider's cloud.
- Wireless Broadband:
 - A 5G tower serves thousands of smartphones.
- Satellite Networks:
 - A single satellite relays data to/from ground stations.

Disadvantages

- Single Point of Failure:
 - If the hub fails, all communication stops.
- Bandwidth Sharing:
 - Remote nodes compete for hub resources (unlike point-to-point's dedicated bandwidth).



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1.11 Network Topologies

1.11.5 Mesh Topology

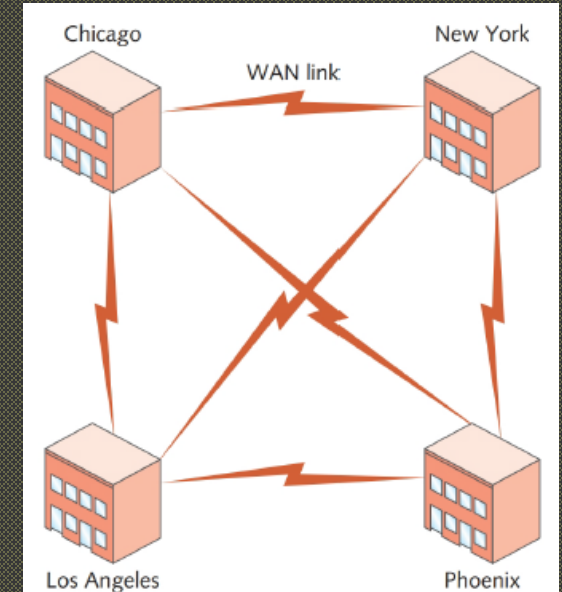
A highly redundant network design where each device (or node) connects directly to every other device, forming multiple point-to-point links.

Types of Mesh Topology:

- Full Mesh:
 - Every device connects to every other device.
 - Maximum redundancy but highest cost.
- Partial Mesh:
 - Devices connect to some (but not all) others.
 - Balances redundancy and cost.

Applications:

- Internet Backbone:
 - Major ISPs interconnect via partial mesh to ensure uptime.
- Military/Government Networks:
 - Full mesh ensures communication survives link failures.
- Data Centers:
 - Servers/switches use partial mesh for low-latency failover.



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1.11 Network Topologies

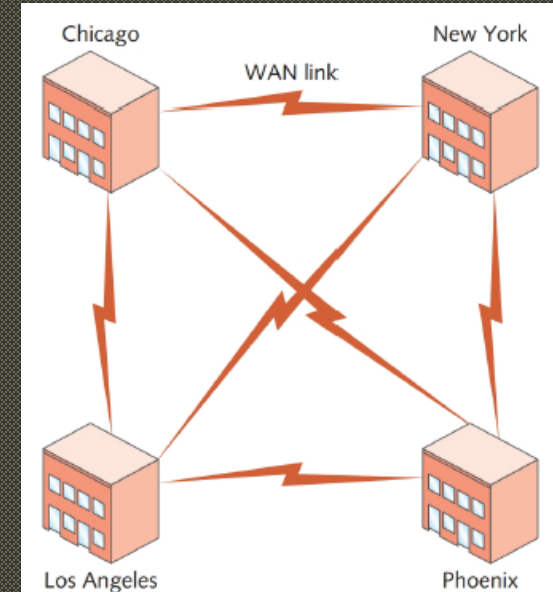
1.11.5 Mesh Topology

Features

- Fault Tolerance:
 - If one (or multiple) links fail, data reroutes through alternate paths.
- High Reliability:
 - Critical for WANs, ISPs, and data centers (e.g., the Internet uses partial mesh).
- Performance:
 - Uses high-speed links (e.g., 10 Gbps fiber) between nodes.

Disadvantages

- High Cost:
 - Requires extra cabling, ports, and infrastructure (e.g., fiber between buildings).
- Complex Maintenance:
 - Managing numerous connections is labor-intensive.



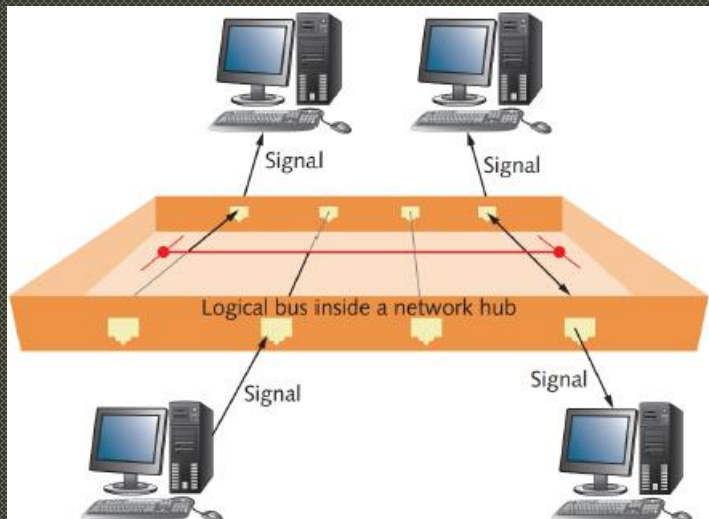
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1.11 Network Topologies

1.11.6 Logical Topology

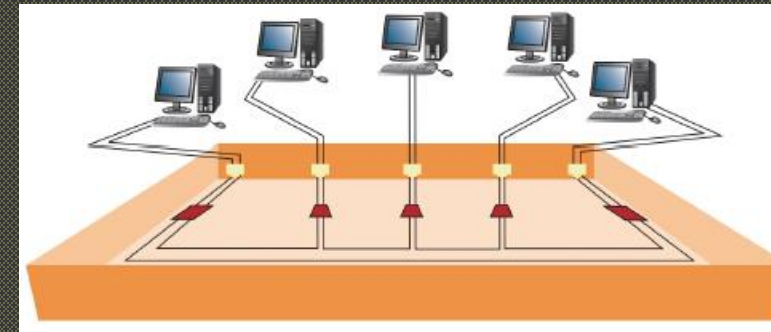
Logical Bus

- Bus: Data travels in all directions along a single cable.
- Star: A hub mimics a bus by broadcasting data to all ports (hidden "logical bus").
- Signals flood the network (like water in pipes).
 - Example: Early Ethernet (10BASE-T with hubs).



Logical Ring

- Star: A token ring MAU (Multi-station Access Unit) creates a virtual ring internally.
- Data passes sequentially between devices via a token (even if physically wired as a star).
- Unused ports are bypassed.



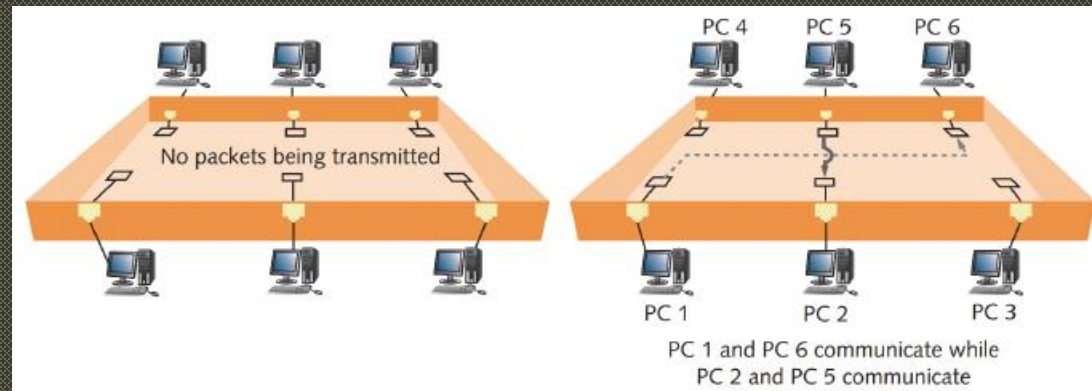
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1.11 Network Topologies

1.11.6 Logical Topology

Switched Topology

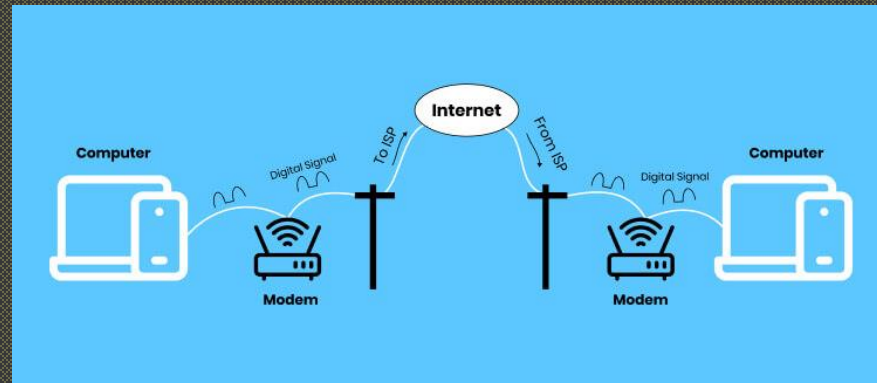
- Physical Star, Logical Direct Connection:
 - No constant circuit: Switches only link devices during data transfer.
 - On-demand paths: A temporary "logical circuit" forms between sender/receiver.
- Advantage:
 - Enables full-duplex, collision-free communication (unlike hubs).



1.11 Network Topologies

1.12 Internet Service Provider (ISP)

- An Internet Service Provider (ISP) is a company that provides access to the internet for individuals, businesses, or other entities. It acts as a gatekeeper to connect a home, office, or device to the internet.
- ISPs manage the infrastructure and services needed for internet connectivity, and they often offer additional services like email, web hosting, or security features.
 - Internet Access: They connect devices (phone, computer, router) to the internet using technologies like fiber, cable, DSL, or wireless.
 - Data Transmission: They ensure data (like web pages, videos, or emails) travels between your device and the internet.
 - Network Management: They maintain the infrastructure (cables, servers, routers) to keep the connection stable and fast.
 - Additional Services: Many ISPs offer extras like email accounts, cloud storage, or cybersecurity tools.
 - Customer Support: They help troubleshoot issues, like when your Wi-Fi stops working or your connection is slow.



1.11 Network Topologies

1.12 Internet Service Provider (ISP)

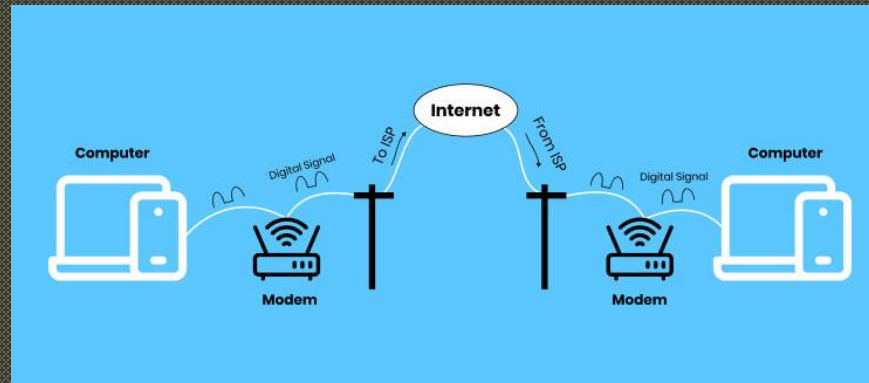
- Explanation of steps for ISP to get customers online:

1. Infrastructure Setup:

- ISPs own or lease physical infrastructure like fiber-optic cables, copper lines, satellite dishes, or cellular towers.
- They connect this infrastructure to the global internet backbone (a network of high-speed cables and data centers that carry internet traffic worldwide).
- Example: Bell lays fiber-optic cables in Toronto to connect homes to its data centers, which link to the global internet.

2. Connection to a Device:

- The ISP provides a connection to a home or business via a modem, router, or other equipment.
- This connection can be wired (e.g., cable, fiber, DSL) or wireless (e.g., 4G/5G, satellite).
- Example: When you sign up with Bell, they install a modem in your home to convert their fiber signal into something your devices can use.



1.11 Network Topologies

1.12 Internet Service Provider (ISP)

3. Data Routing:

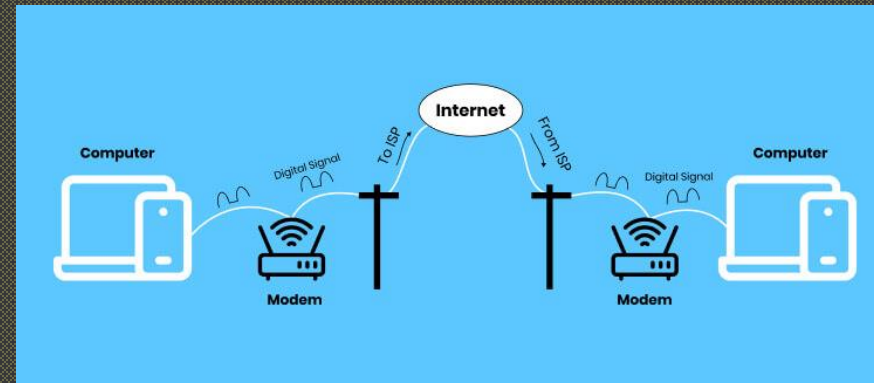
- When customers browse a website or stream a video, their device sends a request through the ISP's network.
- The ISP routes this request to the correct destination (e.g., Netflix's servers) and brings the data (like the video) back to customers.
- Simple Analogy: The ISP is like a postal service, delivering letters (data) to the right address and bringing replies back.

4. IP Address Assignment:

- ISPs assign customer's device an IP address (a unique identifier) so it can communicate on the internet.
- They may use dynamic IPs (which change periodically) or static IPs (fixed, often for businesses).
- Example: When you connect to Rogers's network, they assign your laptop an IP address to track and route your internet traffic.

Challenges with ISPs

- Monopolies: In some areas, only one ISP is available, limiting choices and potentially raising prices.
- Speed Variability: Actual speeds may be lower than advertised, especially during peak hours.
- Data Caps: Some ISPs limit monthly data usage, charging extra if exceeded.
- Privacy Concerns: ISPs can track unencrypted traffic, raising privacy issues unless you use a VPN.



1.11 Network Topologies

1.12 Internet Service Provider (ISP)

5. Bandwidth and Speed Management:

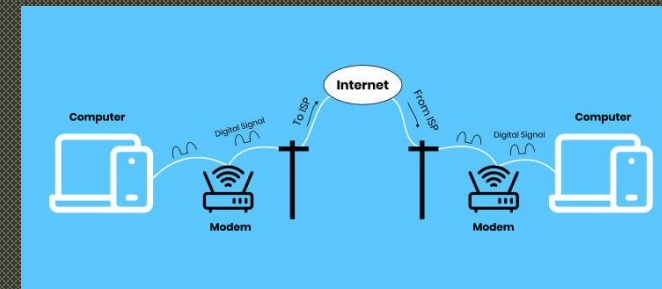
- ISPs control how much data can be sent/received (bandwidth) and the speed of that connection.
- They offer different plans (e.g., 100 Mbps, 1 Gbps) based on specific needs.
- Example: Spectrum might offer a 300 Mbps plan for casual browsing or a 1 Gbps plan for heavy streaming and gaming.

6. Security and Monitoring:

- ISPs may provide basic security features, like blocking malicious websites, or offer advanced tools like VPNs or antivirus software.
- They also monitor network traffic to manage congestion or detect issues.
- Example: Rogers provides a security suite with its internet plans to protect customers from malware.

ISPs deliver internet using different technologies, each with pros and cons:

- Fiber-Optic: Super-fast, reliable, uses light signals through glass fibers.
- Cable: Uses coaxial cables, widely available, good speeds but can slow during peak usage.
- DSL (Digital Subscriber Line): Uses existing phone lines, slower than fiber or cable.
- Wireless (4G/5G): Uses cellular towers, great for mobile or home use in areas without cables.
- Satellite: Works anywhere with a clear sky view, but can have latency issues.



References

- Guide to Networking Essentials, Greg Tomsho, 8th Edition

INDUSTRIAL NETWORKING

MENG 4010

Thank you!

Discussions?